Archives of Computational Methods in Engineering (2023) 30:4401–4448
https://doi.org/10.1007/s11831-023-09942-9

SURVEY ARTICLE

A Comprehensive Survey on Higher Order Neural Networks and Evolutionary Optimization Learning Algorithms in Financial Time Series Forecasting

Sudersan Behera[1] · Sarat Chandra Nayak[2] · A. V. S. Pavan Kumar[1]

Received: 27 February 2023 / Accepted: 5 May 2023 / Published online: 23 May 2023
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Abstract
The financial market volatility has been a focus of study for experts over past decades. While stockbrokers and investors expect reliable projections of future stock indices, it instead displays unpredictable, complicated, and nonlinear reactions which paves path towards designing accurate prediction mechanisms for approximate stock market behaviour. Many computational intelligence methods have been used in the field of economic forecasting alongside more conventional methods. Yet, a trustworthy forecasting depends heavily on the correct selection of an efficient model and optimum set of model hyperparameters. In this regard, artificial neural networks (ANNs) specifically higher-order neural networks (HONNs) have shown notable performance in modelling the chaotic behaviour of financial data and predicting future data. Improved precision and nonlinear decision boundaries are two benefits of using HONN. They can store more data, map features beautifully with weights that are configurable on a single layer, learn quickly, and even tackle complex real-world situations. Besides, evolutionary optimization algorithms (EOA) are widely employed for adjusting HONN parameters thus, enhancing their generalization capability. HONNs are trained with several flavours of evolutionary learning algorithms to discover optimal solutions for various real-world complex engineering problems. The purpose of this article is to explore HONN and EOA based financial forecasts from the last two decades and see how well these forecasts performed in terms of making accurate and reliable financial predictions. We conducted a rigorous survey on state-of-the-art HONN and EOA based methods used for financial forecasting collected from reliable and reputed sources, analysed their applicability and performability, identified their strength and limitations, and suggested few constructive criticisms. Findings of this study may assist researchers of this field in adopting such forecasting methods appropriately.

1 Introduction

Everything that can be banked, invested in the stock market, or transformed into digital currency is considered part of the financial market. Moreover, the stock market's traces have come to represent many financial characteristics; therefore, this development has had a profound effect on the economy. The term "trading" refers to the practise of purchasing and selling financial products such as stocks, equities, commodities, futures, etc. Due to the unpredictable nature of the stock market, investors are exposed to a potentially disastrous environment in which to put their money. If you're an investor, you need to do your homework before making any commitments. Researchers, brokers, investors, and financial institutions are all interested in developing methodologies for accurate stock market prediction because of the stock market's notoriously erratic and unpredictable nature. By extrapolating the current stock price and anticipated stock movement, analysts may estimate what the market will look like in the future and the dangers that come with it. A stock market trend is the general upward or downward price movement of a stock. It is quite challenging to mine the securities market trend due to the market's intensive nonlinearity, high uncleanness, and non-stationary behaviour. As a result

✉ Sarat Chandra Nayak
saratnayak234@gmail.com; saratnayak234@yonsei.ac.kr

Sudersan Behera
sudersan.behera@giet.edu

A. V. S. Pavan Kumar
avspavankumar@giet.edu

[1] Department of Computer Science and Engineering, GIET University, Gunupur, Odisha, India

[2] Department of Computer Science, Yonsei University, Seoul, Republic of Korea

of the seemingly random nature of the securities market, it has become an intriguing topic for stock analysts and scientists to investigate the underlying causes of this phenomenon. Investor sentiment, political decisions, the state of the economy, and the price of commodities all have a profound impact on the behaviour of the securities market. As a result, all investors and their dependents can benefit from a sufficient effort to uncover the underlying dynamics of stock information for accurate prediction.

As a result, conducting research in the field of financial marketing is frequently a difficult task. Both qualitative and quantitative data, such as company profiles, price indexes, volume, etc., are included in the fundamental analysis. Artificial intelligence, machine learning, data mining, opinion mining, deep learning, and tweet analysis have all been extensively used for stock market forecasting. Concerns about feature selection and proper parameter tuning also arise when making use of such a computational approach to forecasting. Models of artificial neural networks (ANNs), including multilayer perceptron (MLP), multiple linear regression (MLR),support vector regression (SVR) [1, 2], recurrent neural network (RNN) [3, 4], convolution neural network (CNN) [5], high dimensional neural networks [6], and other deep learning (DL) [7] methods, arose because of a need to automate the solution of practical problems, such as stock prediction, using computational intelligence [8]. It has been observed from a number of research viewpoints that ANNs occupy a very excellent niche in a number of cutting-edge research subfields, and the advent of HONNs has led to a dramatic rise in the reliability of predictions and the precision of classifications. Researchers lean toward HONNs because of their unique properties, such as large memory capacities, fast computation speeds, easy input–output mapping, and quick learning times. Meanwhile, evolutionary algorithms like genetic algorithm (GA) [9], particle swarm optimization (PSO) [10], monarch butterfly optimization (MBO) [11, 12], [13], teaching–learning based optimization (TLBO) [14, 15],follow the leader (FTL), cuckoo search (CS) [16], harmony search (HS) [17, 18], honey bees mating optimization (HBMO), artificial bees colony (ABC) [19, 20], firefly algorithm (FFA) [21], fireworks algorithm (FWA) [22], grey wolf optimization (GWO) [23, 24], gravitational search algorithm (GSA) [25], ant colony optimization (ACO),chemical reaction optimization (CRO) [26], black hole optimization (BHO) [27], immune algorithm (IA), differential evolution (DE), plant grow optimization (PGO), genetic network programming (GNP) [28], bat algorithm (BA) and social spider optimization (SSO) [29], covariance matrix adaptation evolutionary strategy (CMA-CS) [30], Accelerated gray Wolf Optimization (AGWO) [31] and others are meant to automate the problem-solving skills of nature. Recently, the concept of combining ANN with an evolutionary algorithm has been expanding at a rapid rate. Many other angles on the hybridization concept have been investigated. These include weight training, parameter tuning, architectural design, feature selection, learning rate optimization, ANN structure optimization, etc.

This examination has a dual purpose. An in-depth examination of HONNs' potential for economic forecasting is the first step. The second objective is to demonstrate how efficient evolutionary algorithms are at hybridization, with the best possible response being sought in the area of economic prediction through the use of these algorithms. We test the predictive abilities of HONN and its variants on a number of different financial time series. We opted for HONNs because of their numerous advantages over conventional ANNs when applied to predicting the future values of financial time series. Some of the advantages of HONNs are: faster learning ability [32], stronger approximation capability, larger storage, and higher fault tolerance capability [33], lower computational complexity [34], lower computation overhead, faster convergence [35], improved prediction accuracy [36], nonlinear decision boundaries, therefore offering a better classification capability [37], equipped with one or no hidden layer, good generalization ability, good non-linear mapping capabilities, and being highly adaptable in nature.

EOAs, on the other hand, have the best characteristics that overcome limitations of conventional learning strategies, such as slow convergence and being struck into local minima, so we found they are better suited than traditional learning methods for training ANNs to predict the financial time series. Some of the well-known characteristics are fast convergence, parallelism, being able to find the global optima, necessitating a lesser number of adjustable parameters [33], being more robust with fewer tunable parameters and having good approximation capability [38], being intuitive, having ease of implementation, and having the ability to effectively solve highly nonlinear mixed integer optimization problems that are typical of complex engineering systems [39]. Some other advantages of EOAs are that they are capable of solving both single and multi-objective optimization problems, they do not strike in local minima, they are easy to use, they are capable enough to perform global search, they have greater exploration and exploitation capabilities, they can capture the non-linearity in time series data, they have a high ability to determine both optimal and hyper-optimal parameters of the network model, etc.

As a result, it would be wise to conduct a comprehensive survey in order to fulfil the aforementioned prerequisites. Two research-based questions are addressed in this survey to achieve these aims (RBQs).

Research based query1 (RBQ1): How accurate are HONNs at predicting financial time series?

Research based query2 (RBQ2): How good are evolutionary algorithms when it comes to teaching ANNs to make accurate financial time series predictions?

With a specific application to financial market forecasting in mind, this research provides a systematic framework for investigating HONNs and various evolutionary optimization strategies. Figure 1 provides a schematic summary of the whole study. In the portions of this survey devoted to each subset, possible answers to the formulated RBQs are discussed.

1.1 Motivation

One of the most promising areas of study in the scientific world is artificial neural networks. Recently, a number of studies have been conducted by researchers on the applications of ANNs and DL techniques in various domains, including cryptocurrency prediction [40], stock prediction [28, 41, 42], exchange rate prediction [42, 43], crude oil [42, 44], quaternionic domain [6], image processing [45], inventory management [46], estimation of virtual power plants [47], time series forecasting [48], spectrum sensing [49–51], routing [49], airways disorder illness [52], and solar energy forecasting [53]. While numerous studies have shown success using ANNs like MLP for classification and prediction, the model has been shown to have drawbacks like local minima, sluggish convergence, and excessive computation time. Researchers have shown interest in HONNs [54] to get around these shortcomings of MLP. HONN has been employed in many prediction and classification issues due to its many advantageous characteristics, such as its short learning time, good representation of input space, and great learning skills that require less memory in terms of loads and nodes. HONN have a number of appealing characteristics, including quick learning, a large store capacity, fault tolerance, and superior mapping. Because of the higher-order expressions with which it is associated, HONN is now capable of providing superior nonlinear boundaries and high classification accuracy compared to conventional linear neurons. Evolutionary optimization, on the other hand, is a more complex strategy. Optimization entails looking for the best possible way to achieve a desired goal. Changes to input signals, fine-tuning of device or simulation model parameters, and iterative mathematical exploration are only a few of the numerous activities involved in the optimization process. Metaheuristic algorithms, of which evolutionary algorithms are a subset, may be further classified into two subsets: those that use a population-based approach and those that use a single population.

Additionally, we may state that evolutionary algorithms are metaheuristics that rely on a population. An abundance of metaheuristic algorithms [55] draws ideas from natural systems. Further, many optimization issues may be effectively resolved by using evolutionary algorithms. Here, we undertake a comprehensive literature review of HONN-based financial time series forecasting. The primary reasons we chose HONNs for this specific survey are their high quality and reliability. Our selection of an evolutionary-based optimization technique stems in large part from our interest in investigating the efficacy of various EOAs in training HONNs across several dimensions. Various research studies have proven financial prediction using evolutionary computing. Advances in evolutionary computation and hybridization of evolutionary approaches with ANNs especially with

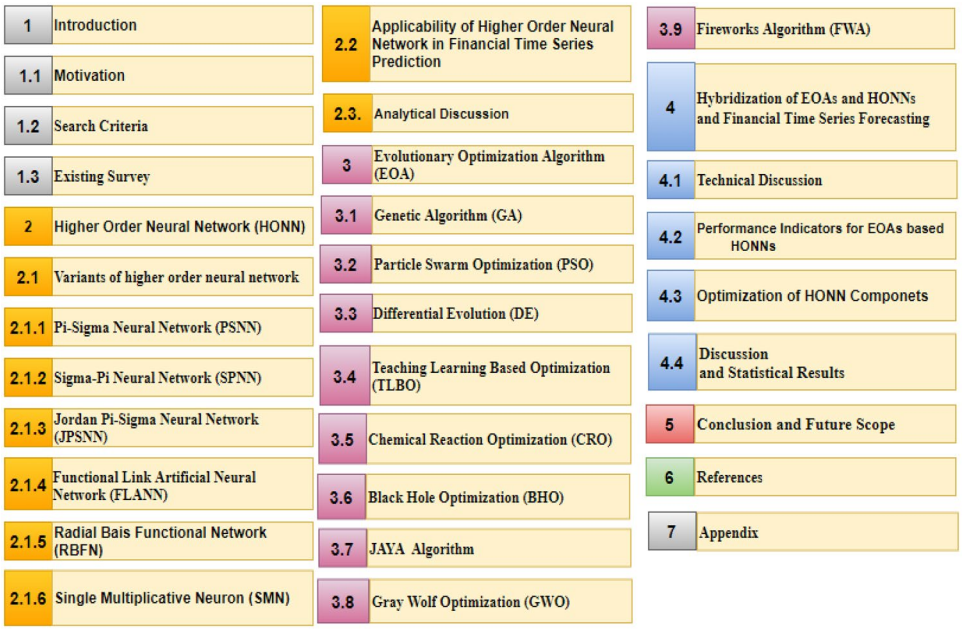


Fig. 1 Organisation of our article

HONNs for financial market forecasting require complete research. In this article, we give a broad overview with HONNs as the main focus. We talk about the different ways that evolutionary computation has been used to predict index values, crude oil prices, gold prices, cryptocurrencies, exchange rates, and stock trends, among other things.

1.2 Search Criteria

Our hope is that this overview will be comprehensive enough to include HONN applications in economic forecasting. We used an innovative search strategy to compile a significant number of published publications that were most related to our survey's primary purpose, allowing us to perform a credible study. Using search terms like "financial market prediction + sigma-pi neural network (SPNN)", "stock market prediction + pi-sigma neural network (PSNN)", "stock market prediction + functional link artificial neural network(FLANN)", "stock market prediction + recurrent neural network (RNN)", "financial time series forecasting + Jordan SPNN", "stock market prediction + single layer neural network (SLNN)", "financial time series prediction + single layer feed forward network (SLFN)" 'evolutionary algorithm + stock market prediction", "EOA + financial time series prediction", "financial time series forecasting + HONN + EOA", "Metaheuristic based optimization + financial time series forecasting", "financial time series forecasting + second order neural network (SONN)", "financial time series forecasting + condensed polynomial neural network (CPNN)", financial time series forecasting + echo state network (ESN) etc. we compiled some of the articles from Google Scholar, where some of the articles are downloaded. To make sure the most recent developments could be categorised together, we gathered all the articles published between 2007 and 2023 against our search criteria and then we filtered them to find out the most relevant articles which are suitable for our survey. Figure 2 presents a pie chart which displays the volume of articles collected year wise which are most relevant as far as our survey is concerned.

After segregating all the collected articles year wise, we thoroughly categorized them publisher wise and Fig. 3 contains a bar chart which presents publisher wise volume of articles collected.

1.3 Existing Survey

According to a review of HONN conducted by [54], different implementations of HONN are used for tasks as diverse as prediction, classification, optimization, and function approximation. However, the financial prediction arena ignores HONN's most important uses. They also found that some evolutionary algorithms are combined with few variants of HONN to boost the hybrid model's efficiency. A comprehensive study of PSNN [56] was conducted, documenting the network's efficacy in a wide range of applications, including function approximation, temperature forecasting, categorization, and prediction. The results of their investigation reveal that PSNN and JPSNN, a variation of PSNN, play key roles in a variety of contexts. Similarly, a comprehensive survey only on PSNN for time series prediction was conducted in [57]. The present survey focuses heavily on the categorization aspect, and only a few HONN versions are employed. There is still a need to learn how HONN might help with financial forecasts. On the other hand, other scholars have undertaken surveys on EOAs, but they are focused

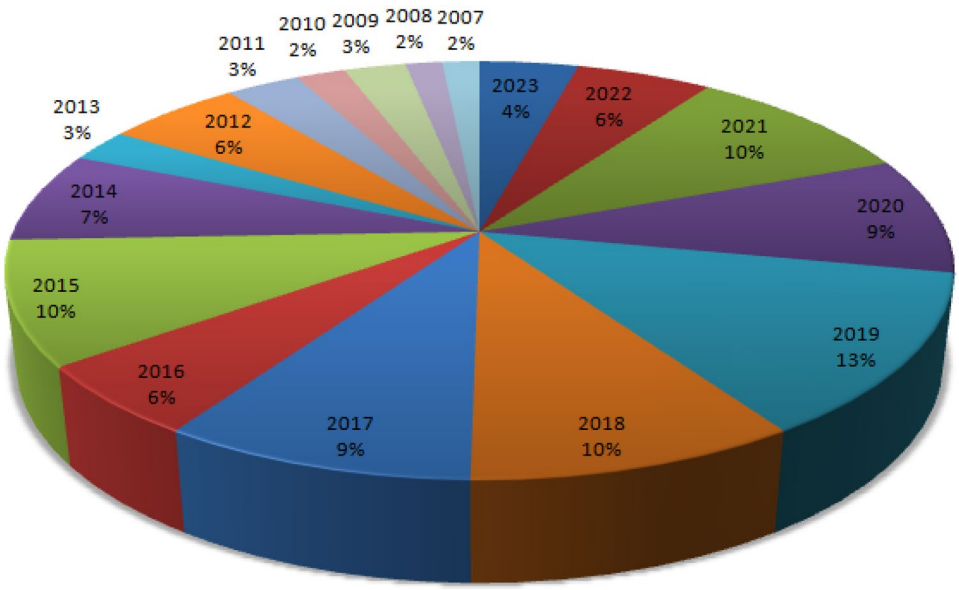


Fig. 2 Year wise most relevant number of articles collected as far as our search criteria concerned

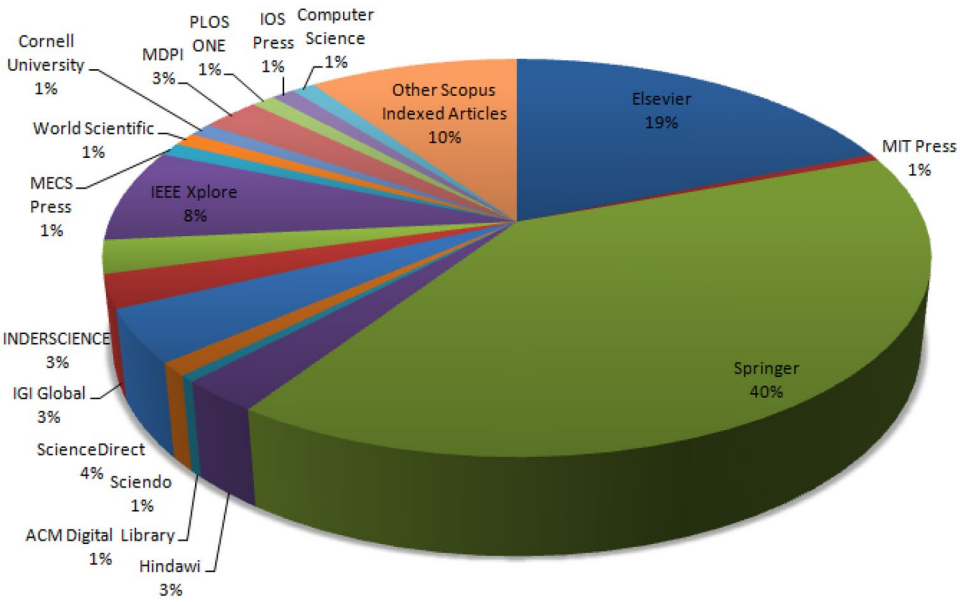


Fig. 3 Publisher wise number of articles collected for our survey

on certain fields such as newly proposed EOAs [58, 59], function optimisation [59], hyper-parameters optimization [60], intelligent system development [61], phase equilibrium thermodynamic calculations [62], bioinformatics [63], algorithmic trading [64], classification [65], prediction, and so on. However, little emphasis has been placed on the usefulness of EOAs in the prediction of financial time series. There was also no survey that focused on the combined effect of EOAs and HONNs on financial time series forecasting. In this study, we mainly concentrated on the effective applications of HONN, especially in financial time series forecasting. On the other hand, various evolutionary algorithms have been investigated in order to accumulate the effective contributions of EOA in hybridization with ANNs, especially HONNs for time series prediction.

2 Higher Order Neural Network (HONN)

Normally an HONN contains zero or one hidden layer and one changeable weight. HONN variations and operation methods vary. Some do summation and then multiplication, while others do the opposite. HONN quickly learns nonlinearity due to a single layer of teachable loads. Higher words boost the HONN's representational power in solving complicated problems and information capacity. HONNs are excellent for researchers due to their simple structure and high computing ability.

2.1 Variants of Higher Order Neural Network

Considering the presence or absence of hidden layer, order of operations, input signal expansions etc. HONN got different variants. Some of the well known HONNs are PSNN, SPNN, JSPNN, FLANN, RPNN, DRPNN, SONN, SLFN, ELM, CPNN, ESN, SMNM, RBFN, FLN and name a few. Due to simple structure, it is easy to get a variant of any HONN for better results. Some of the frequent use HONN models are briefly covered in this section.

2.1.1 Pi-Sigma Neural Network (PSNN)

A PSNN has three layers. Between the input and middle layers is a weight set. The intermediate layer is a concealed unit that performs summing, although its load is unity [54]. Calculation layer calculates model output. Weighted inputs are connected to the hidden layer, which performs summing, and the result passes to the output layer's product unit. Only one adjustable hidden layer reduces this network's learning time. This network's product unit gives higher order capabilities and stronger nonlinear separability through exploration without exponential growth of weights. Figure 4 depicts the basic layout of a PSNN for prediction.

The product of the weighted input signal between the *ith* input unit and the *jth* hidden unit is the output of the *jth* summation unit, as shown in Eq. 1.

$$Y_{jh} = \sum_{i=1}^{N} w_{ij} \times x_i \tag{1}$$

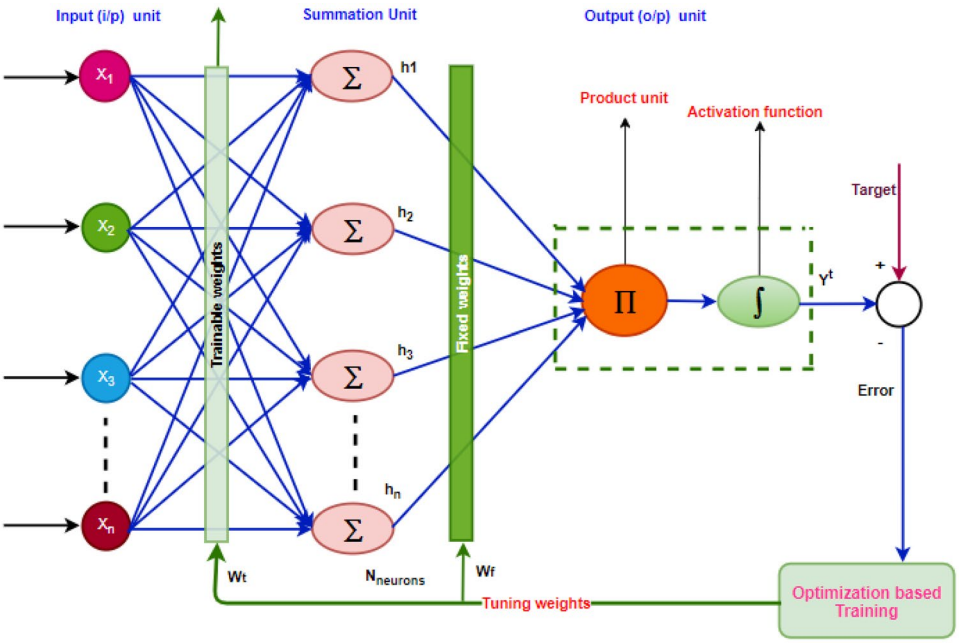


Fig. 4 Pi-sigma neural network architecture for prediction

By simplifying, we can write this as $N =$ the sum of all input signals, $x_i =$ the signal from *ith* input unit, and $w_{ij} =$ the weight between *ith* and *jth* hidden units. Here's how to get the network's output from the information produced by the *jth* hidden unit:

$$O_{psnn} = \sigma\left(\prod_{j=1}^{k} y_{jh}\right) \tag{2}$$

In Eq. 2, k represents the network degree and σ is an activation function. The network degree is the number of hidden nodes. The contribution of PSNN in financial time series forecasting is discovered in the literature [14, 19, 21, 32, 33, 35, 37, 66–90] according to our survey. Similarly, the application of different PSNN variants in financial time series can be found in literature [91–96].

2.1.2 Sigma–Pi Neural Network (SPNN)

A SPNN employs the sum of input signal components [67]. In the initial step of computing, it multiplies weighted input signals, and the results pass through a summation unit and activation function. Figure 5 shows the SPNN model architecture for prediction. The use of SPNN in financial time series prediction may be found in the literature [67, 97]. Similarly, the use of a modified version of SPNN in financial time series can be observed in [98].

Equation 3 is used to determine the output at the *jth* unit of the hidden layer.

$$Y_{jh} = \prod_{i=1}^{n} w_{ij} \times x_i \tag{3}$$

The output layer's final value can be computed using the following equation: Eq. 4.

$$O_{spnn} = \sigma\left(\sum_{j=1}^{k} y_{jh}\right) \tag{4}$$

2.1.3 Jordan Pi–Sigma Neural Network (JPSNN)

In a JPSNN, RNN capabilities are superimposed on a PSNN's structure. It features a unique output-to-input feedback connection [54]. Together, the time-series data and this setup regulate the network's temporal dynamics and make for a more consistent and efficient flow of operations. The architecture of the JSPNN model is depicted in Fig. 6. A significant application of JPSNN in financial time series forecasting is found in literature [67, 99–101].

If at time T the signal at each of the *kth* external inputs is z_k, the load is w_{ij}, the hidden layer summing unit is $h_k(T+1)$, and the output is $Y(T+1)$ at time $T+1$. After that, we can use Eq. 5 to determine the JSPNN's final result:

$$Y(T) = \begin{Bmatrix} z_k(T) & if, 1 \le k \le N \\ 1 & if, k = N+1 \\ o_k(T) & if, k = N+2 \end{Bmatrix} \tag{5}$$

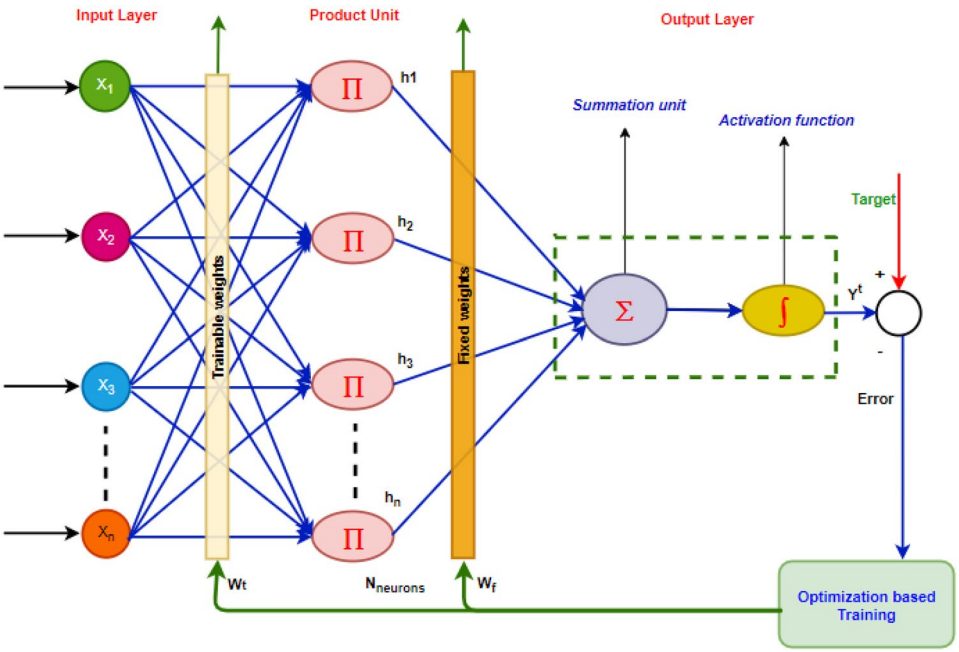


Fig. 5 Sigma-pi neural network architecture for prediction

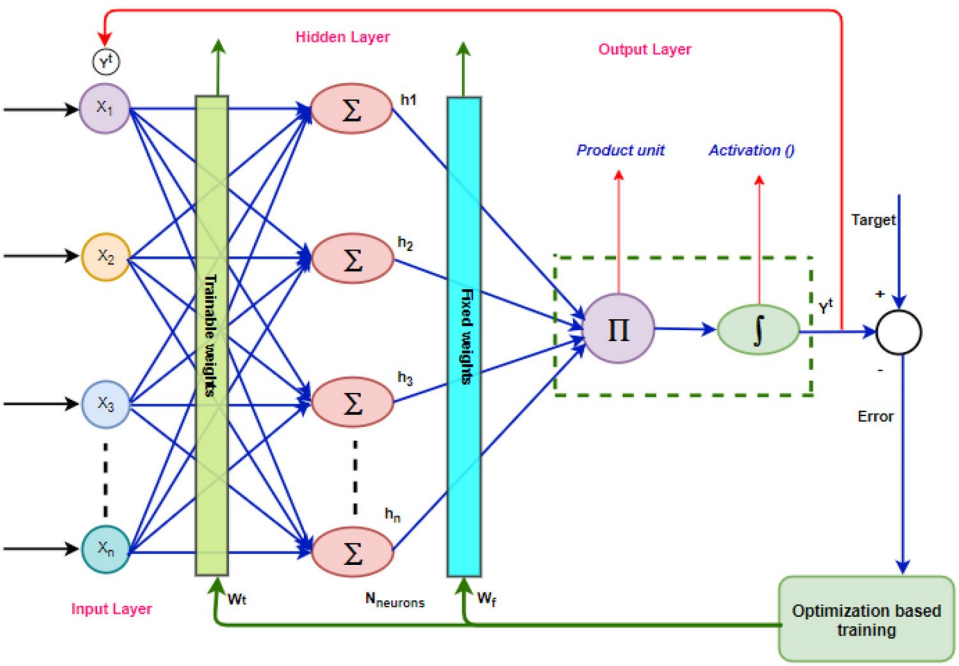


Fig. 6 Jordon pi-sigma neural network architecture for prediction

2.1.4 Functional Link Artificial Neural Network (FLANN)

FLANN does not have a hidden layer like other HONNs [67]. Due to no hidden layers, it's straightforward and delivers a decent approximation. The input signal is extended through a functional expansion unit. This approach creates a hyper plane with improved discrimination. To map time series irregularities, extra functions can be added to the function expansion unit, and in some circumstances, specialised functions should be employed to obtain the desired result. It's flat and computationally efficient [102]. Figure 7 shows the FLANN model architecture for prediction. In our survey, contribution of FLANN in financial time series forecasting is found in literature [17, 34, 69, 70, 79, 80, 84, 103–114]. Similar to FLANN, the following FLANN variations have been applied to forecasting financial time series: HFLANN [115–117], CFLANN [55, 118, 119], LFLANN [55, 118,

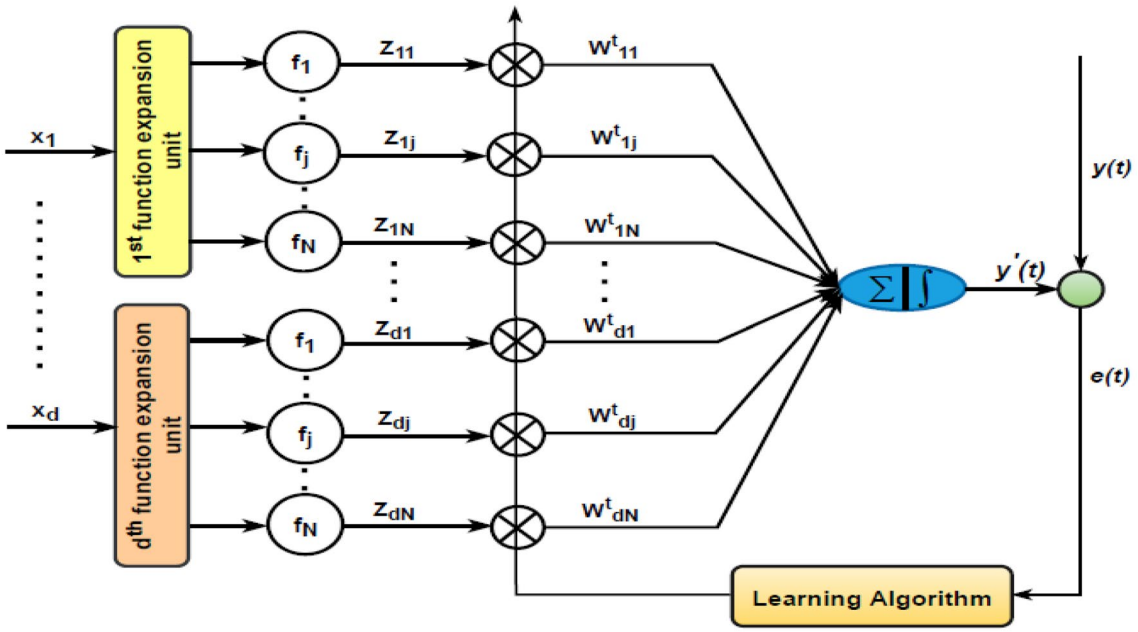


Fig. 7 Functional link artificial neural network architecture for prediction

119], PFLANN [55], TFLANN [55, 118, 119], LeFLANN [118, 119], TPFLANN [120], FFLANN [121], RCE-FLANN [122, 123], RFLANN [124], and FLN [125–128].

The expansion of input signal depends on the type function used in the functional expansion unit and it varies function to function. If x_i is the input signal and Legendre polynomial is used for the input expansion then the input pattern expansion can be given as follows:

$$Z_1(x_i) = x,\ Z_2(x_i) = \frac{3x^2 - 1}{2},\ Z_3(x_i) = \frac{5x^3 - 3x}{2},\ Z_4(x_i) = \frac{35x^4 - 30x^2 + 3}{8}$$

In addition to this the higher order polynomial can be calculated as follows:

$$Z_{n+1}(x_i) = \frac{1}{n+1} * ((2n+1) * x_i * Z_N(x_i) - nZ_{n-1}(x_i)) \quad (6)$$

The output of the network can be written as follows:

$$y'(t_1) = \sigma\left(\sum_{k=t-p}^{t-1}\left(\sum_{m=1}^{N}(z_k(m) \times w_k^j(m)\right) + w^j(0)\right) \quad (7)$$

and the error can be calculated as follows:

$$e^j(t_1) = y(t_1) - y'(t_1) \quad (8)$$

here y' is the predicted output, y is the actual output, σ is an activation function e^j is the error of the network, $w^j(0)$ represents the bias at jth iteration for the input $X(t_1)$.

2.1.5 Radial Basis Function Network (RBFN)

RBFN is an efficient tool for both pattern recognition and function approximation [129]. It consists of two layers of network nodes. The weighted sum of the hidden units is then sent to the output unit, which calculates the output using the same radial activation function as is used in the computation of the hidden units [129]. It is a subclass of HONN in which the activation of hidden neurons is determined by the offset between the input and prototype vectors, denoted by r_i and p_i is the input space patterns, or vectors, form clusters, and p_i stands for their centres. Between the middle layer (the hidden layer) and the output layer, a weighted connection was set up. The model architecture of the RBFN is shown in Fig. 8. Interested reader may find the application of RBFN in financial time series prediction in literature [130–133]. Similarly, applications of some modified RBFN in financial time series research can be found in literature [134] and [135].

The output of the RBFN is the linear combination of hidden unit signal and given in Eq. (9) as follows:

$$y^t = f(c,x) = \sum_{i=1}^{N} c_i h_i(x) \quad (9)$$

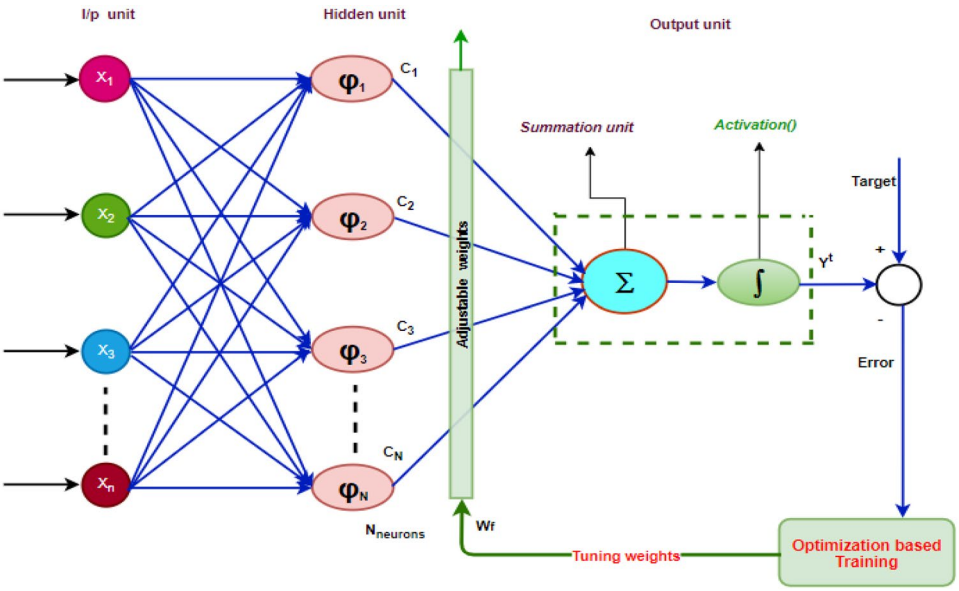


Fig. 8 Radial basis function network architecture for prediction

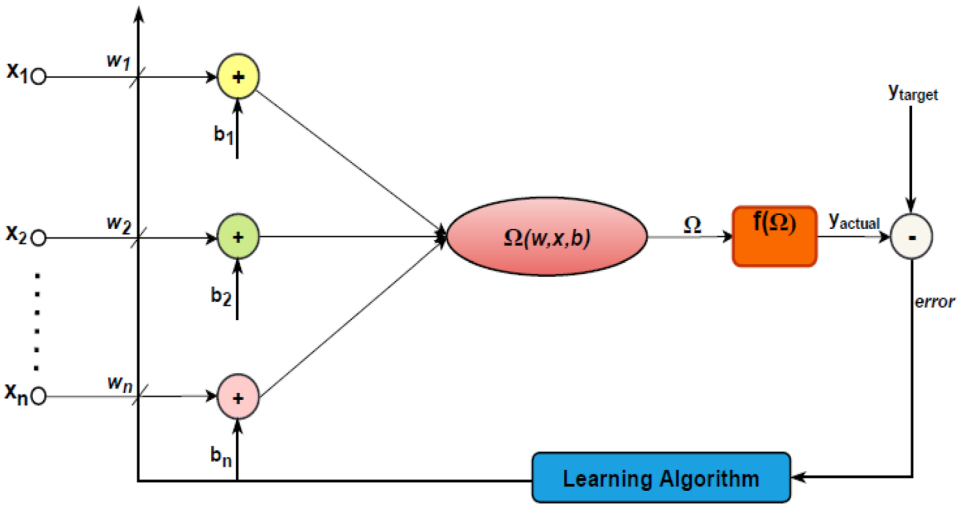


Fig. 9 Single Multiplicative Neuron model for prediction

$$h_i(x) = \exp\left(-\sum_{k=1}^{m}(x_k - p_{ik})^2 / r_{ik}^2\right) \tag{10}$$

where p_i is the centre of a region and r_i is the width of receptive field.

2.1.6 Single Multiplicative Neuron (SMN)

For function approximation and other purposes, the artificial single multiplicative neuron model was first developed by Yadav et al. [136]. Using Vapnik's statistical learning theory, they examine the model's capabilities, including its nonlinear generalisation, input–output mapping, and noise tolerance. For tasks that normally need a multi-layered neural network, the single-neuron model should be adequate. The model and the learning algorithm are provided in the following sections. The model architecture of SMN is shown in Fig. 9.

For $i \in n$, the network's input, weight, and bias are denoted by x_i, w_i and b_i respectively. As seen in Eq. (11), the operator performs a multiplicative action.

$$\Omega = \prod_{i}^{n}(w_i x_i + b_i) \tag{11}$$

The output function is the *logsig* function defined as Eq. (12)

$$y = \frac{1}{1 + e^{-\Omega}} \tag{12}$$

The learning algorithm is used to minimize the error between the output of the model and actual values. The

application of SMN model in financial time series forecasting can be found in [137, 138].

2.2 Applicability of HONNs for Financial Time Series Forecasting

Our study concentrates on the exceptional performance of HONNs in financial market prediction, although they are also used in many other financial applications. HONNs and their variants are heavily used for financial time series predictions because of their low complexity structure, faster learning properties, and better generalization power, and they are found to be more reliable and stable models than other ANN models. To invest profits, traders may need a reliable prediction strategy and clear laws. Building a reliable prediction model is tough due to non-linearity and market volatility. A reliable financial prediction model is error-tolerant, capable of capturing market anomalies and uncertainty, and produces high accuracy by minimizing or completely eliminating mistakes. Another trait of a reliable prediction model is its capacity to adapt to any training procedure in order to get the best outcome.

Nayak et al. [66] used a PSNN to forecast stock index prices and found it outperformed other network models with an APE of 0.853. Dash et al. [32] proposed a PSNN to forecast GOLD/INR and GOLD/AED exchange rates, which outperforms in computing time, RMSE, MSE, and MAE. For instance, Nayak [33] used a PSNN to make accurate predictions about the future cost of crude oil. The experimental findings demonstrated the superiority of the model created by Nayak et al. [35] to forecast the index of the HIS, DJIA, BSE, and KOSPI stock markets using a PSNN. Dash et al. [68] presented a PSNN and found it increased the accuracy of their predictions for the USD/CHF and USD/JPY exchange rates. The PSNN model given by Nayak et al. [21] was found to be a stable, reliable classifier with better accuracy. Yilmaz et al. [37] used a pi-sigma artificial neural network (PS-ANN) to predict the AUST and TEC time series data, and their results were superior to those of other methods. Patnaik et al. [74] used a PSNN to predict fuzzy time series data and found that it was more accurate than others. Nayak and Naik [75] proposed a PSNN classifier that was stable, dependable, and accurate. Kanungo et al. [76] found the PSNN to be a more accurate classifier. Dash [71] presented a PSNN to anticipate USD/CAD/CHF/JPY exchange rates. The proposed model outperformed and predicted well. Using a PSNN, Patnaik et al. [72] were able to predict the fuzzy time series. Sahoo et al. [73] used a PSNN to predict BSE stock prices and found good performance for tiny outliers. Nayak et al. [14] used a PSNN classifier and found it converged faster and was accurate. Nayak et al. [94] used AE-PSNN, an adaptive evolutionary higher-order network, to anticipate stock market index fluctuations. The model outperformed standard HONNs. Nayak et al. [78] found that PSNN was a good way to classify real-time series data from the UCI machine learning repository. Samir et al. [83] proposed an improved pi-sigma neural network (IPSNN) for non-linear system control, which enhanced outcomes. Nayak et al. [77] suggested a statistically significant PSNN classifier. Bas et al. [81] used a PSNN to get higher-order fuzzy time series that could predict fuzzy relationships and improve performance. Nayak et al. [86] suggested a PSNN model and evaluated it with five benchmark datasets from the UCI machine learning library. For all datasets, the new model did better than the others and gave IRIS datasets the most accurate results. Nie & Deng [87] improved function optimization using a PSNN.

Dash et al. [88] built a PSNN model and tested it on two datasets, gold/INR and gold/AED, to predict future gold prices. The model outperformed MSE, RMSE, and MAE. Husaini et al. [82, 89] presented a PSNN higher-order neural network to predict Batu Pahat suburban temperatures. PSNN consistently beats MLP in all metrics. Sahu et al. [90] used a PSNN to predict foreign exchange rates more accurately than previous models. A. J. Hussain et al. [95] tested a recurrent pi-sigma neural network (RPSN) on two physical time series datasets. The suggested model outperforms the feed-forward pi–sigma network by 1.85 dB. Akram et al. [96] used three physical time series datasets to test an improved PSNN with error feedback (PSNN-EF). They found that it could be used instead of PSNN and JSPNN. Yilmaz et al. [85] tested a pi-sigma artificial neural network (PS-ANN) using AUST and TEC time series. The suggested model outperformed existing ANN models. Akdeniz et al. [92] showed an ARMA-type pi-sigma artificial neural network (ARMATPS-ANN) for non-linear time series forecasting and got better RMSE and MAPE values. Sahu et al. [55] suggested four FLANN variants—PFLANN, CFLANN, LFLANN, and TFLANN—to estimate foreign exchange rates. PFLANN and LFLANN performed better in most situations.

Nayak [67] constructed hybrid models using four HONNs—PSNN, SPNN, JPSNN, and FLANN to forecast exchange rates. Das et al. [102] used FLANN to forecast stock closing prices and found that accuracy increased with reduced computational cost. Sahu et al. [34] used a FLANN (FLN) to predict the time series of currency exchange rates and found that it converged faster and did better. Das et al. [118] used a poly-FLANN to predict the BSE, NASDAQ, FTSE, TAIEX, and DJIA daily closing indices. After exploring and adding VDPs to the supplied data, they applied functional expansion. The suggested model beats the MLP and FLANN models. Husaini et al. [79, 80] developed two HONNs, PSNN and FLANN, for physical time series forecasting. Both models outperformed other models for all datasets. Naik et al. [103] used a FLANN classifier to solve the classification challenge, with successful results. Nayak

and Ansari [69] used PSNN and FLANN for stock forecasting with good results. Sahu et al. [91] used a PSNN as a base model and constructed three evolving HONNs (EHONN) to model and predict five exchange rate time series. The suggested model is a promising exchange rate forecasting tool. Waheeb and Ghazali [120] made a genetically optimized tensor product functional link neural network (TPFLNN) for predicting daily exchange rates. It had a structure with less complexity, learned faster, and was more accurate than other FLNNs and traditional models. Soumya et al. [17] trained a FLANN for stock market prediction and found that it outperformed the other two models in execution time and RMSE.

Anish and Majhi [121] created a feedback FLANN (FFLANN) with recursive least squares (RLS) training for stock market prediction. It was more accurate than previous models. Rout et al. [122, 123] used a recurrent computationally efficient FLANN (RCE-FLANN) to forecast S&P 500 and NIKKEI 225 stock prices and got good results. Nayak et al. [125] developed an artificial functional link network (ACFLN) model to predict stock prices and tested it on the DJIA, BSE, TAIEX, FTSE, and NASDAQ. Finally, the suggested model handled stock market risks better than FLANN and standard models. The FLANN model was built for classification purposes by B. Naik et al. [104–106] and evaluated on many datasets retrieved from the UCI machine learning library. They compared the suggested model to others and found it to be superior. Nayak et al. [124] found that for predicting financial time series, recurrent FLANN (RFLANN) did better than other models. Yogi et al. [107] found FLANN channel equalization acceptable for non-linear and non-fading channels. Dehuri [108] used FLANN to classify problems and found that it overcomes non-linearity better than SVM, FLANN, and RBF. The FLANN model predicts the Indian stock market better than other models, according to Mahapatra et al. [109]. Nayak et al. [127] developed an FLN for stock market forecasting that outperformed previous models. Patra et al. [112] made a FLANN for predicting stock prices that was faster to compute than MLP. R. Rodriguez [139] used a Multiscale Legendre neural network (MLNN) with multi-scale stationary wavelet decomposition to make better predictions about anchovies every month. Nayak et al. [119] introduced four variations of FLANN for stock index forecasting: TFLANN, LeFLANN, CFLANN, and LFLANN. LeFLANN worked well on all data sets because it was simple to use and had a flat structure.

Nayak [126] used four polynomials for FLN input signal expansion to predict bitcoin closing prices. TFLN and LeFLN, which captured Bitcoin's hidden dynamics, had the greatest forecast accuracy. Nanda et al. [110] employed two robust single-layer ANNs, a FLANN and a Lagurre ANN (LaNN), to anticipate the foreign exchange rate. FLANN and LaNN outperformed ARIMA, although LaNN had superior accuracy and convergence. Nanda and Tripathy [111] presented a polynomial neural network (PNN), Legendre neural network (LeNN), and FLANN to forecast opencast mining equipment noise. FLANN outperformed. Luo et al. [36] used a deep feed forward neural network (DFNN) to estimate the amount of electricity used more accurately and in less time. Dash [140] made DECPNN, a model that uses both DE and Chebyshev polynomial neural networks (CPNN) to predict stock prices. Dash and Dash [141] found that Legendre neural networks (LENNs) forecast financial time series with greater APE, accuracy, and computational complexity without losing speed. Nayak & Misra [38] used an Extreme Learning Machine with Chemical Reaction Optimization (ELCRO) to forecast stock volatility. The ELCRO is statistically significant, has better MAPE and ARV values, and converges more quickly. DRPNN was developed by Ghazali et al. [141–145] to forecast stationary and non-stationary financial time series. They showed that DRPNN has faster convergence, less network complexity, less prediction error, faster learning, and better performance to catch the chaotic signal movement and make more profit.

Wang et al. [146] created a general regression neural network (GRNN) and trained it using a form of PSO to forecast the TWD/USD and RMB exchange rates. With minimal elapsed time, GRNN predicted better. Sudeepa et al. [147] used an optimized extreme learning machine and a cluster-based modified crow search algorithm (CMCSA) to forecast a random nonlinear stock market price (OELM). Intrinsic Time-Scale Decomposition (ITD) was used to decompose non-stationary stock price data into proper rotation components (PRCs) and a residual component. OLEM outperformed other competitive models. Zhang and Hu [148] used an ultra-high-frequency sigmoid and cosine artificial higher-order neural network (UGC-HONN) to predict the USD/CAD and USD/AUD exchange rates. This network was better than other HONNs by 2.8546 to 3.4185%. Egrioglu et al. [149] tested a median pi-artificial neural network (MdPNN) on the CIF-2016 dataset. MdPNN outperforms MdANN, TrANN, and PSNN in predicting, according to experiments. Nayak & Misra [150] introduced a condensed polynomial neural network (CPNN) to anticipate stock closing price indices. CPNN beats previous approaches and is a promising forecasting model. Nayak et al. [151, 152] used a second-order neural network (ASONN) as a HONN to predict financial time series and found it outperformed other state-of-the-art models. Husaini et al. [101] used a Jordan pi-sigma neural network (JPSNN) to find that JPSNN was better at predicting the temperature than both MLP and PSNN.

For classification tasks, Nayak et al. [99, 100] suggested a JPSNN and found that it improved accuracy. Kumar [131] suggested three models for predicting intraday stock prices: an RBFNN, a time-delayed NN (TDNN), and a BPNN. The BPNN worked best, he found. When Prasetyo et al. [153] used a HONN to predict how much oil would be produced

in the future; they found that the suggested HONN gave more accurate results. Dash [154] used a recurrent Legendre polynomial neural network (RLPNN) to predict the FOREX market and found that it was more accurate than other prediction models. After implementing two HONNs models, including FLNN and PSNN, for chaotic time series forecasting, Waheeb and Ghazali [84] found that both models outperformed several other commonly used hybrids. Wang et al. [155] have used an echo state network (ESN) for time series prediction and found that the model has better accuracy for small sample sets than others because of its powerful nonlinear capacity and effective and uncomplicated training techniques. For optimization purposes, Zapata [156] used an ESN and found a significant decrease in computing expenses. Chouikhi et al. [157] employed an ESN for forecasting time series and discovered that the generalization has improved. Ghazali et al. [158] used a ridge polynomial neural network (RPNN) for multi-step-ahead financial time series prediction and found that RPNN yields greater profit returns with faster convergence than FLANN, PSNN, and MLP due to its rapid learning and strong mapping capabilities.

For financial time series forecasting, Waheeb et al. [159] propose a dynamic RPNN (DRPNN) with a lyapunov function, and they find that DRPNN performs well in terms of RMSE and learning speed. Jumeily et al. [160] used a DRPNN for predicting physical time series and said that it performed better than MLP, FLANN, and PSNN. Much focus has been placed on the network's output by Waheeb et al. [161], who created a recurrent probabilistic neural network (RPNN) with feedback (RPNN-EF) and analyzed its efficacy on four univariate time series. In the end, they discovered that, compared to RPNN, RMSE improved by an average of 23.34% when using RPNN-EF and by an average of 10.74% when using DRPNN-EF. Arsian and Yolcu [97] used a hybrid SPNN (HSPNN) to predict intuitively fuzzy time series, and it did a better job than other cutting-edge prediction methods. Akbilgic et al. [132] successfully utilized a hybrid RBFNN for financial research. For chaotic time series prediction. Aladag et al. [162] used a multiplicative seasonal neural network (MSANN) with excellent results. Dehuri and Cho [117] showed that HFLNN was a better way to classify things than FLNN and RBFN. Decai et al. [163] used a robust echo-state network (RESN) to predict chaotic time series and found that it could handle outliers and give good results. The work of Zhong et al. [164], who used a double-reservoir echo state network (DRESN) for multi-regime time series prediction, shows that the suggested model works and does a better job than the standard models.

Egrioglu et al. [165] suggested a recurrent multiplicative neuron model ANN (RMNM-ANN) for non-linear time series forecasting and showed that it outperformed other models. By using a Legendre neural network (LeNN), Nanda et al. [166] were able to improve upon previous regression models for forecasting air quality. Xiong et al. [134] made a fully complex-valued radial basis function neural network (FCRBFNN) as a possible way to predict interval time series. The forecasting performance has increased, as noted by Bas [167], who suggested a multiplicative neuron model artificial neural network (MNM-ANN) for financial time series forecasting. Yolcu et al. [138] developed a single MNM-ANN with an autoregressive coefficient (AC-S-MNM) for time series modeling, and they found that this model outperformed all others by a wide margin. Zhao and Yang [137] created a single multiplicative neuron model (SMNM) for financial forecasting. They later proved that their SMNM worked and was reliable. Das et al. [168], who claim that extreme learning machines (ELMs) are well-suited for a model predicting the next day's stock closing price, have proposed ELMs for financial prediction. Dehuri and Cho [115] developed a hybrid FLANN for classification tasks and discovered that the HFLANN outperformed the RBFN and FLANN and readily handled the non-linearity present in classification challenges. For fuzzy time series forecasting, Egrioglu et al. [19] used a PS-ANN with successful outcomes. Mili and Hamdi [116] found that a hybrid version of the FLANN (HFLANN) did a better job of classification than the FLANN. To improve upon the performance of existing models for fuzzy time series forecasting, Kocak et al. [93] suggested an ARMA-type pi-sigma ANN (ARMATPS-ANN). When Rather et al. [169] integrated RNN, ES, and ARMA to produce HPM for stock return prediction, they discovered excellent prediction performance had been obtained and the proposed model outperformed the RNN model.

The use of a feed-forward neural network (FNN) improved the accuracy of currency exchange rate forecasts by Worasucheep et al. [170]. Majhi et al. [129] made a multiobjective evolutionary algorithm based on RBFN and found that it did a better job of predicting stock prices than traditional RBF. Nayak and Misra [171] used a neuro-fuzzy hybrid network (CRO-NFN) based on chemical reaction optimization to predict stock prices at market close. Smith and Jin [3] used an evolutionary multi-objective creation of recurrent neural network ensembles to make predictions about time series and found that RNN did better than other models. To forecast the closing price and volatility of five major stock indices, Dash et al. [130] used a hybrid learning framework to train two single hidden layers feed forward neural networks (SLFNs), i.e., RBFs, and a low complexity functional link artificial neural network (CEFLANN). After evaluating both models, they concluded that CEFLANN was superior to RBF. When using the FLANN classifier that Dash et al. [114] described, there have been reports of improved accuracy for inconsistent datasets. Shrivastava and

Verma [135] made a cascaded SOM-RBF (CSR) to improve the accuracy of their option price forecasts. To anticipate software failure, Behera et al. [113] used fuzzy logic analysis and a neural network (FLANN), with positive results. In order to capture the uncertainties in exchange rate data, Sahu et al. [172] presented a multiplicative FLANN (MFLN) for exchange rate forecasting. Chang et al. [173] made an evolving partly connected neural network (EPCNN) to predict stock price trends and found that it can accurately predict the stock price index for most data. Tang et al. [174] used an evolutionary pruning neural network (EPNN) model to study financial bankruptcy. They found that it was more accurate and faster to converge than MLP and PNN models. Yang & Duan [175] developed an extreme learning machine (ELM) for financial time series prediction. Nayak et al. [176] trained an ANN and pushed it as a good way to predict the volatility of the financial market. Nayak et al. [177] came up with the idea of an ANN to improve the accuracy of financial research.

Chen et al. [178] said that an HNN could be used to predict how a river would flow downstream, and their predictions were right. Cao [179] suggested a FNN model for predicting chaotic time series, and the results were excellent. Asadi et al. [180] developed a pre-processed evolutionary Levenberg–Marquardt neural networks (PELMNN) model for stock market prediction using genetic algorithms, feed forward neural networks, data pre-processing, and post-processing concepts. The PELMNN model can handle stock market fluctuations and has good prediction accuracy. Nayak et al. [181] developed a chemical reaction optimization algorithm-based dendritic neuron model (DNM) for intelligent financial forecasting and found it to be effective. Cascaded adaptive nonlinear functional link networks (CCFLNN) were used by Nayak et al. [182] to model and predict Crude Oil Prices Time Series, and it was discovered that the suggested model was well-suited for forecasting. Using deep reinforcement learning (DRL) to perform reliable spectrum sensing in the face of SSDF attacks in cognitive radio networks, Paul et al. [50] found improvements over both traditional and SVM-based methods. To improve the robustness of cooperative spectrum sensing against spectrum sensing and data falsification attacks, Paul and Choi [51] proposed an ensemble learning technique based on convolutional neural network (CNN) and deep neural network (DNN) algorithms. Table 1 shows how HONNs and its different versions have been used to make predictions about financial time series.

2.3 Analytical Discussion

It is clear from the reviewed literature (from Table 1) that HONN and its variants are widely used in financial prediction, such as stock index prediction, bitcoin price prediction, wind energy forecasting, function regularization, classification, electricity consumption, and many more. High accuracy, rapid convergence, and low error rates in financial time series prediction are all hallmarks of HONN and its derivatives. One may see how many HONN variations are making useful contributions to financial time series forecasting by looking at the literature. Figure 10 gives a clear picture of the different kinds of HONNs that are used to predict financial time series.

In contrast to the HONNs, whose weights tend to grow exponentially with network size, the PSNN structure requires fewer weights while still being able to solve the prediction problem quickly and accurately. It is clear that PSNN [14, 19, 21, 32, 33, 35, 37, 66–69, 71–90] has been used in the vast majority of studies. PSNN has a straightforward structure, a single trainable weight, and a short learning curve, all of which contribute to its widespread popularity. Our research shows that compared to other hybrid models, PSNN performs better in several aspects, including accuracy, training time, steadiness, reliability, capacity to map non-linearity, and convergence. PSNN is also outperforming some classic predators, including MLP, MLR, and other traditional models. Even PSNN's offspring, such as JPSNN [67, 98–101], AE-PSNN [94], RPSNN [95], PSNN-EF [96], ARMATPS-ANN [92, 93], EPSNN [91] and MSPSNN [98] are better than more traditional models. FLANN [17, 34, 67, 69, 79, 80, 84, 102–114] is a recent development of the progress of a higher-order network that has investigated various effective ways to forecast the major complex financial time series. In light of the foregoing, it may be concluded that a FLANN model performs better than its competitors. It yields more reliable predictions, requires less computing time, yields meaningful statistical data, converges more quickly, and produces fewer errors. There was also a noticeable improvement in performance metrics. The FLANN model has also been compared to some of the more common forecasting models, such as the MLP, ARIMA, BPNN, SVM, SVR, BPRBF, and LSE, and found to be superior in terms of error reduction, precision, and speed of convergence, to name a few. Similarly, many variants of FLANN, i.e., HFLANN [115–117], CFLANN [55, 118, 119], LFLANN [55, 118, 119], PFLANN [55], TFLANN [55, 118, 119], LeFLANN [118, 119], TPFLANN [120], FFLANN [121], RCE-FLANN [122, 123], RFLANN [124], FLN [125–128], and MFLN [172], outperformed both FLANN and other traditional models in terms of flat network structure, low complexity, fast learning, nonlinear mapping, generalization, and high accuracy. HONNs' use of multivariate polynomials, which leads to an explosion of weights, makes them difficult to work with, while RPNN's use of univariate polynomials [158, 161], which makes it simple, makes it a good choice. Despite RPNN's ability to increase information, it aids in solving more complex problems with quick learning. The RPNN can incorporate dynamic systems by taking input and

Table 1 Applicability of HONNs for efficient financial time series prediction

Author and reference	HONN model	Application area	Observations
Nayak et al. [66]	PSNN	Prediction	Best APE value achieved, PSNN found superior than others
Dash et al. [32]	PSNN	Prediction	Good results achieved with less computation time
Nayak [33]	PSNN	Prediction	Superior to other
Nayak et al. [35]	PSNN	Prediction	Superior to others
Dash et al. [68]	PSNN	Prediction	Improved accuracy
Nayak at el. [21]	PSNN	Classification	Fat, steady, reliable model and better accuracy
Yılmaz et al. [37]	PS-ANN (PSNN)	Forecasting	Performance is better than other models
Patnaik et al. [74]	PSNN	Forecasting	Better accuracy over other models
Nayak and Naik [75]	PSNN	Classification	Steady, reliable and better accuracy
Kanungo et al. [76]	PSNN	Classification	Better accuracy
Dash [71]	PSNN	Prediction	Outperformed and found as a promising predictor model
Patnaik et al. [72]	PSNN	Forecasting	Good results achieved
Sahoo et al. [73]	PSNN	Prediction	Performance is better in case of small outlier
Nayak et al. [14]	PSNN	Classification	Faster convergence and high accuracy
Nayak et al. [94]	AE-PSNN	Prediction	Found to be better than class HONN
Nayak et al. [78]	PSNN	Classification	Promising performance for classification
Samir et al. [83]	PSNN	System control	Better results found
Nayak et al. [77]	PSNN	Classification	Model is Statistically significant
Chen et al. [178]	HNN	Forecasting	Good results achieved
Cao [179]	FNN	Prediction	Performed well comparing to another model
Shrivastava and Verma [135]	Cascaded SOM-RBF (CSR)	Prediction	Better prediction accuracy achieved compare to the deep neural network
Bas et al. [81]	PSNN	Prediction	Better performance achieved
Feng et al. [98]	MSPSNN	Optimization	Behaves better than traditional SPSNN
Nayak et al. [86]	PSNN	Classification	Best result of 96.76310% accuracy than the others
Nie and Deng [87]	PSNN	Optimization	Better convergence achieved
Dash et al. [88]	PSNN	Prediction	Better prediction achieved
Husaini et al. [82]	PSNN	Forecasting	PSNN forecast was fashionably outperformed the MLP in all measuring criteria
Sahu et al. [90]	FLANN	Prediction	Efficient Prediction than other
Husaini et al. [89]	PSNN	Prediction	Reasonably better prediction achieved compared to the MLP
Hussain et al. [95]	RPSNN	Prediction	An average improvement of SNR 1.85 dB found over the FFPSNN
Akram et al. [96]	PSNN-EF	Prediction	Lower prediction error than other network models
Yilmaz et al. [85]	PS-ANN (PSNN)	Forecasting	PS-ANN outperformed many ANN models available in the literature
Arsian and Yolcu [97]	Hybrid SPNN	Prediction	The proposed model outperformed to some other state-of-the-art prediction tools
Akdeniz et al. [92]	ARMATPS-ANN	Prediction	Better RMSE and MAPE values achieved
Sahu et al. [55]	FLANN Variants	Prediction	PFLANN and LFLANN perform better in most of the cases
S.C. Nayak [67]	HONN Variants	Prediction	All variants outperformed to traditional models
Das et al. [102]	FLANN	Prediction	Improved accuracy and less computation time
Naik et al. [103]	FLANN	Classification	Superior performance than other
Sahu et al. [34]	FLANN	prediction	Faster convergence, Success over others
Das et al. [118]	Poly-FLANN	Prediction	Poly-FLANN forecasting model generates lowest MAPE values for all datasets
Nayak et al. [194]	ANN	Prediction	Lowest MAPE & ARV values
Worasucheep et al. [170]	FNN	Forecasting	Better forecasting results achieved

Table 1 (continued)

Author and reference	HONN model	Application area	Observations
Waheeb and Ghazali [120]	TPFLANN	Prediction	Low complexity structure, faster and good accuracy
Nayak and Ansari [69]	PSNN, FLANN	Prediction	Both the models are found to be significant and improved accuracy
Sahu et al. [91]	EPSNN	Prediction	Can be a promising forecasting tools for exchange rate forecasting
Dash et al. [130]	RBF, CEFLANN	Prediction	CEFLANN outperformed all other models
Soumya et al. [17]	FLANN	Prediction	RMSE value very lea and faster convergence
Anish and Majhi [121]	FFLANN	Prediction	Produces better accuracy than other models
Rout et al. [122]	RCE-FLANN	Prediction	Produces satisfactory prediction results
Nayak et al. [125]	ACFLN	Prediction	Outperformed to FLANN-GA, MLP-GA, MLP-CRO, RBFNN, and others
Naik et al. [104]	FLANN	Classification	The proposed model is better and outperforms other alternatives
Naik et al. [105]	FLANN	Classification	An average accuracy of 89.2211% achieved
Nayak et al. [124]	RFLANN	Prediction	Produces better accuracy than other models
Rout et al. [123]	RCE-FLANN	Prediction	Superior accuracy achieved
Naik et al. [106]	FLANN	Classification	The proposed Method performs relatively better than other models
Sahu et al. [172]	MFLN	Prediction	Quite capable of capturing the uncertainties associated with exchange rate data
Dehuri et al. [108]	FLANN	Classification	FLANN overcomes the non-linear problems comparing to other models
Mahapatra et al. [109]	FLANN	Prediction	The proposed model found to be superior to other models
Nayak et al. [127]	FLN	Forecasting	The performance gain of the proposed model is superior than other models
Patra et al. [112]	FLANN	Prediction	FLANN is computationally efficient than MLP
Rodriguez [139]	MLNN	Forecasting	Outperformed to MLP and LNN
Nayak et al. [119]	FLANN variants	Prediction	With less complex structure, LeFLANN achieved good average performance
Behera and Behera [196]	RPNN	Classification	Better accuracy achieved
Nayak et al. [176]	ANN	Forecasting	Efficient model for forecasting the volatile data
Nayak et al. [177]	ANN	Prediction	Accuracy is improved
Nayak, [126]	FLN Variants	Prediction	Best forecast accuracies achieved by TFLN and LeFLN
Nanda et al. [110]	FLANN, LaNN	Prediction	LaNN has better accuracy and high convergence than FLANN
Nanda & Tripathy [111]	FLANN, PNN, LeNN	Prediction	FLANN model gives better noise prediction than the PPN and LeNN model
Luo et al. [36]	DFNN	Prediction	Better prediction accuracy with less computation time seen by optimal DFNN
Dash [140]	CPNN	Forecasting	Excellent results in terms of MAPE and RMSE values
Dash and Dash [141]	LENN	Prediction	The computational complexity has reduced
Nayak and Misra [38]	SLFN	Prediction	Statistically Significant, Faster convergence, Superior MAPE and ARV values
Wang et al. [146]	GRNN	Prediction	Better prediction achieved with low elapsed time
Sudeepa et al. [147]	OELM	Prediction	Found to be superior model and accuracy improved
Zhang and Hu [148]	UGC-HONN	Forecasting	UGC-HONN models are 2.8546% to 3.4185% better than other HONNs
Egrioglu et al. [149]	MdPNN	Prediction	Outperformed to MdANN, TrANN and PSNN
Nayak and Misra [150]	CPNN	Prediction	Outperformed over other models
Nayak et al. [152]	ASONN	Prediction	Better forecasting accuracy over other state of the art models
Nayak et al. [151]	ASONN	Prediction	Better forecasting accuracy over other state of the art models

Table 1 (continued)

Author and reference	HONN model	Application area	Observations
Husaini et al. [101]	JPSNN	Prediction	Lower prediction error, JPSNN found to be superior to MLP and PSNN
Nayak et al. [99]	JPSNN	Classification	Performance is better in case of small outlier
Nayak et al. [100]	JPSNN	Classification	Better accuracy achieved
Kumar [131]	RBF, TDNN, BPNN	Forecasting	BPNN achieved higher accuracy
Nayak et al. [181]	DNM	Prediction	Able to capture the underlying uncertainties coupled with financial data
Dash et al. [114]	FLANN	Classification	Better accuracy has found for inconsistent datasets
Lu et al. [128]	ESN	Prediction	Better results found
Paul et al. [50]	DRL	spectrum sensing	Better results achieved than conventional and Support Vector Machine-based approaches
Prasetyo et al. [153]	HONN	Prediction	Better prediction accuracy
Husaini et al. [79]	PSNN, FLANN	Prediction	Accuracy improved to 0.707% by PSNN and 0.670% by FLANN
Husaini et al. [80]	PSNN, FLANN	Prediction	Outperformed in terms of lower error
Dash [154]	RLPNN	Prediction	Found effective prediction model than others
Waheeb and Ghazali [84]	FLNN, PSNN	Forecasting	FLNN and PSNN offer good performance than others
Wang et al. [155]	ESN	Prediction	Reliable prediction and enhance accuracy
Ghazali et al. [158]	RPNN	Prediction	Fast learning, powerful mapping, fast convergence and higher profits seen
Ghazali et al. [143]	DRPNN	Prediction	Reduced network complexity, smaller prediction error and faster learning
Waheeb et al. [161]	RPNN-EF	Forecasting	Outperformed to RPNN and DRPNN
Waheeb et al. [159]	DRPNN	Forecasting	An average 21.45% of improvement in RMSE and learning speed
Ghazali et al. [142]	DRPNN	Prediction	Improvement in the profit return and rapid convergence over others
Ghazali et al. [144]	DRPNN	Prediction	Improvement in the profit return, and rapid convergence over MLP
Jumeily et al. [160]	DRPNN	Prediction	DRPNNs produced improved SNR, and superior to others
Zapata [156]	ESN	Optimization	Larger prediction horizons, computational costs were greatly reduced
Akbilgic et al. [132]	Hybrid RBFNN	Forecasting	HRBF-NN model is a highly flexible
Aladag et al. [162]	MSAAN	Forecasting	High forecasting accuracy achieved
Egrioglu et al. [165]	(RMNM-ANN)	Forecasting	The proposed model produced the most accurate forecasts
Dehuri and Cho [117]	HFLNN	Classification	Performed well over FLNN & RBFN
Chouikhi et al. [157]	ESN	Prediction	Simple and powerful network, Generalization has improved
Tang et al. [174]	EPNN	Prediction	Superior to others in terms of accuracy, convergence speed, etc
Yang and Duan [175]	ELM	Prediction	Superior performance and competitive computational efficiency
Behera et al. [113]	FLANN	Prediction	Better results achieved
Paul and Choi [51]	CNN and DNN	Joint spectrum sensing	Proposed work outperformed the existing works found in the literature
Decai et al. [163]	RESN	Prediction	RESN found to be robustness to outliers and provides satisfactory prediction accuracy
Zhong et al. [164]	DRESN	Prediction	DRESN found to be effective in predicting multi-regime time series
Nanda et al. [166]	LeNN	Prediction	LeNN has good prediction capacity as compared to regression models
Xiong et al. [134]	FCRBFNN	Forecasting	FCRBFNN is a promising alternative model for interval time series forecasting
Bas [167]	MNM-ANN	Forecasting	Forecasting performance improved

Table 1 (continued)

Author and reference	HONN model	Application area	Observations
Yolcu et al. [138]	AC-S-MNM	Forecasting	AC-S-MNM exhibits significantly better performance than other
Zhao and Yang [137]	SMNM	Prediction	The propose model is found to be efficient and robust
Das et al. [168]	ELM	Prediction	ELM model can be considered as a suitable tool to predict next day closing price
Dehuri and Cho [115]	HFLANN	Classification	HFLANN overcomes the nonlinearity problems and outperformed to other
Egrioglu et al. [19]	PS-ANN	Forecasting	Best forecasts results obtained and suggested suitable for multivariate time series
Mili and Hamdi, [116]	HFLANN	Classification	Faster convergence and better performance than FLANN-BP
Kocak et al. [93]	ARMATPS-ANN	Forecasting	Better forecasting performance achieved
Rather et al. [169]	HPM	Prediction	Outstanding prediction performance has been obtained
Nayak et al. [182]	CCFLNN	Prediction	The proposed model is found to be suitable for forecasting
Asadi et al. [180]	PELMNN	Prediction	Can be considered as a promising alternative for stock market prediction
Majhi et al. [129]	RBFN	Prediction	Prediction performance superior to that of the conventional RBF
Nayak and Misra [171]	NFN	Prediction	Significantly better performance than other models
Ghazali et al. [145]	DRPNN	Forecasting	DRPNN outperforms others due to its extremely nonlinear dynamical nature
Smith and Jin [3]	RNN	Prediction	Better than other models
Chang et al. [173]	EPCNN	Forecasting	A promising tool in forecasting of the financial time series data
Yogi et al. [107]	FLANN	Channel Equalization	The proposed equalizer is found suitable for non-linear and non-fading channels
Paul et al.[49]	SVM	Prediction	Simulation results show the efficacy of the present work over the conventional approach

output history into account, but this representation is not conducive to a network with feedback loops. But on the contrary, due to the recurrent feedback in a DRPNN [142–144, 159, 160], the network outputs are contingent not only on current and past inputs but also on those from outside the system. DRPNN's memories are capable of overcoming the aforementioned restriction and displaying excellent dynamic performance. Other variants of HONN like EPNN [174], RLPNN [154], MSANN [162], RESN [163], DRESN [164], SMNM [136, 137], AC-S-MNM [138], MNM-ANN [167], RMNM-ANN [165], RNN [3, 4] and its variants [3, 146, 169], EPCNN [181], ELM [168, 175, 183], OLEM [130, 147] and PELMNM [180] are also shown effective and better prediction results than conventional models. RBFN [80, 129–132], FCRBFNN [134], SOM-RBF [135] also shows significant improvements in time series forecasting comparing to other traditional models. In comparison to standard HONNs and ANNs, the error rate and accuracy of predictions made by other hybrid HONNs were significantly better. The success of HONN and its variants in predicting financial time series can be inferred from our discussion so far. **(RBQ1).**

We also looked at the popularity of HONNs from the perspective of researchers. Figure 11 makes it clear how frequently researchers use HONNs to forecast financial time series.

We also looked at how academics in other fields test their own models for making predictions by using financial time series datasets. We found several financial data sets that academics have used to test the performance of the financial prediction and classification. To simplify analysis and presentation, we classified those datasets. We carefully removed duplicate datasets in each category. Each group's datasets are given below:

Stock market data = [BSE, DJIA, NASDAQ, TAIEX, FTSE, HIS, NIKKEI, BSE SENSEX, CNX NIFTY, NIKKEI225, S&P 500, KOSPI, NSE, NIFTY, HIS, IXIC, N100, NSEI, RUT, GDAXI, Russell2000, ICICI Prudential, Birla, TAIFEX, and BIST100], Exchange rate = [GBP/USD, YEN/USD, INR/USD, EUR/USD, AUD/USD, GBP/USD, USD/AUD, USD/GBP, USD/INR, USD/EUR, USD/JPY, USD/CAD, USD/AUD, Gold/INR, Gold/AED, UK/EU, US/EU, JP/EU, JP/UK, UK/EU, JP/UK, PKR/USD, BDT/USD, CNY/USD, NPR/USD, LKR/USD, USD/CHF, and USD/

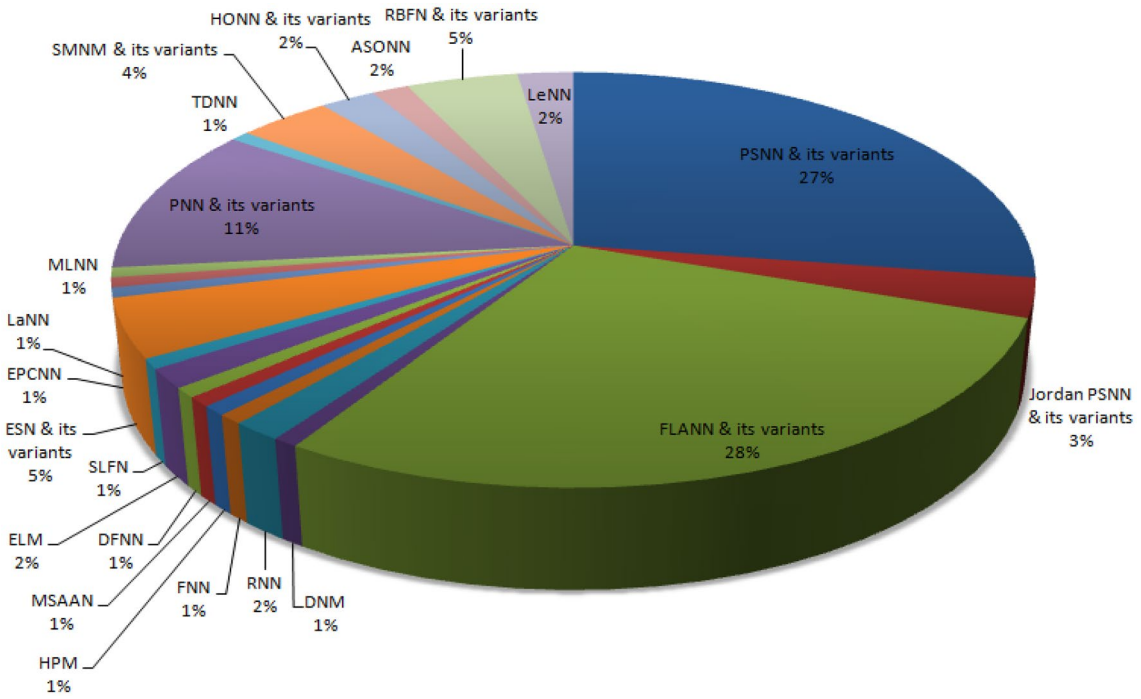


Fig. 10 Types of higher-order networks used in financial time series prediction

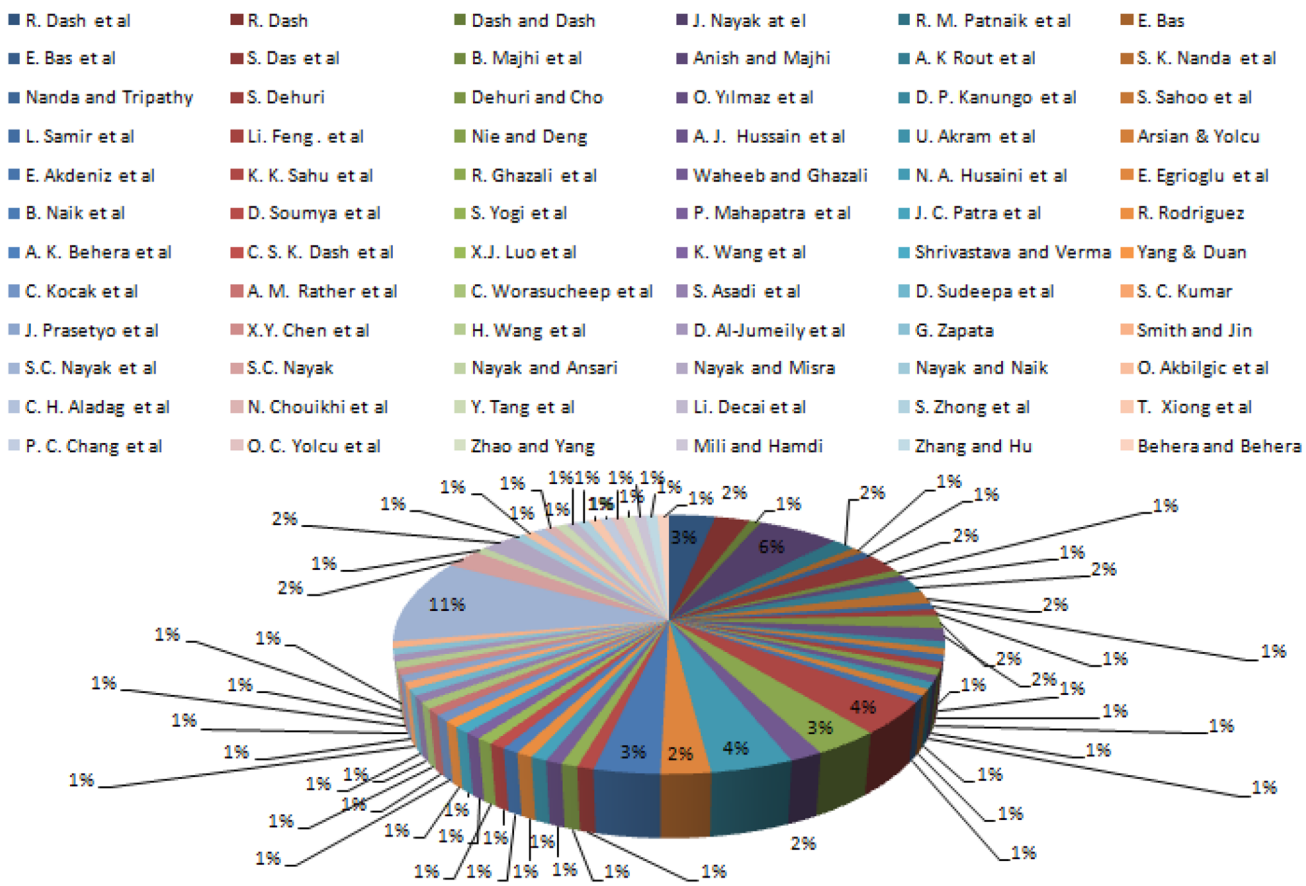


Fig. 11 Frequent uses of HONNs by researchers for financial time series prediction

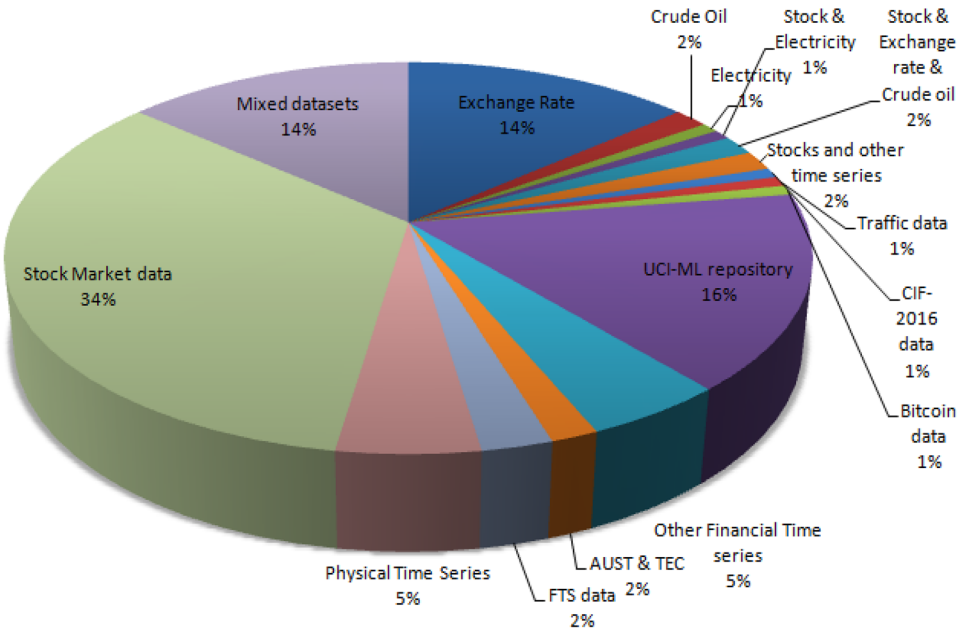


Fig. 12 Data sets used for financial time series prediction

YEN], Crude Oil = [crude oil data sets], Stock and electricity = [S&P 500, NIKKEI 225, and PJM electricity statistics], Stock, Exchange, and Crude Oil = [IBM, USD/EUR, Brent crude oil, NASDAQ, EURO/USD, crude oil, BIT/USD], CIF-2016 data, Traffic data = {Lozi, Tent, SR29-S-402863 dataset], AUST & TEC = [AUST & TEC time series data],Stocks and other time series = [Lorenz attractor, AE index, temperature, and sunspot data], Bitcoin = [Current Bitcoin prices], UCI-ML repository = [UCI machine learning repository],Physical Time Series = [Relative Humidity, Santa Fe Laser, NO2, CO, SO2, and O3, Evaporation, Wind, and Humidity, and Temperature Data Sets of the Malaysian Meteorological Department (MMD)], Other Financial Time Series = [Mackey–Glass, Henon, Lorenz, S&P 500, Magnetic Levitation, Joint Angles, NARMA time series, Rossler benchmark, Yellow River annual runoff; EEG data; Box-Jenkins], Mixed datasets = [UCI machine learning repository and exchange rate data, bankruptcy dataset and distress dataset, Mackey–Glass, star brightness, sunspot data, EUR/USD, CB1, CB2, time series data datasets, Temp, Ozone, Gold, Bitcoin, Pima Indians Diabetes Database, Sonar data, Turbofan engine multi-regime time series, and machinery noise data from Mahanadi Coalfields Limited (MCL), Talcher (Orissa)], Electricity = [Electricity consumption] and FTS data = [Fuzzy time series].

A variety of financial statistics fall into many categories, as shown in Fig. 12. These datasets are then utilized to make predictions about the future of the financial sector.

3 Evolutionary Optimization Algorithm (EOA)

Due to their population-based design and inspiration from natural selection and evolution, evolutionary algorithms are classified as a type of metaheuristic. Researchers frequently employ evolutionary techniques to fine-tune their models, functions, and methodologies in search of the best possible answer to a research question. Numerous researchers have spent a lot of time probing the waters of numerous applications of evolutionary optimization techniques, including but not limited to forecasting, classification, optimization, regularisation, training, prediction, etc. In every scenario, evolutionary optimization methods outperform their more conventional counterparts. Error minimization, nonlinearity mapping, accuracy, and other areas have all been proven to benefit from evolutionary approaches. Since they are based on observations of nature, new evolutionary optimization methods are bound to emerge over time. In this study, we have gone over some of the many evolutionary optimization methods that have made useful contributions to the field of monetary time series evaluation. We have reviewed several of the most well-known evolutionary optimization techniques, including GA, PSO, and DE, in this section. We also provided a brief elaboration of a few newly proposed evolutionary metaheuristics, including TLBO, CRO, BHO, JAYA, GWO, and FWA. By reading the academic literature, interested readers can learn more about the remaining EOAs.

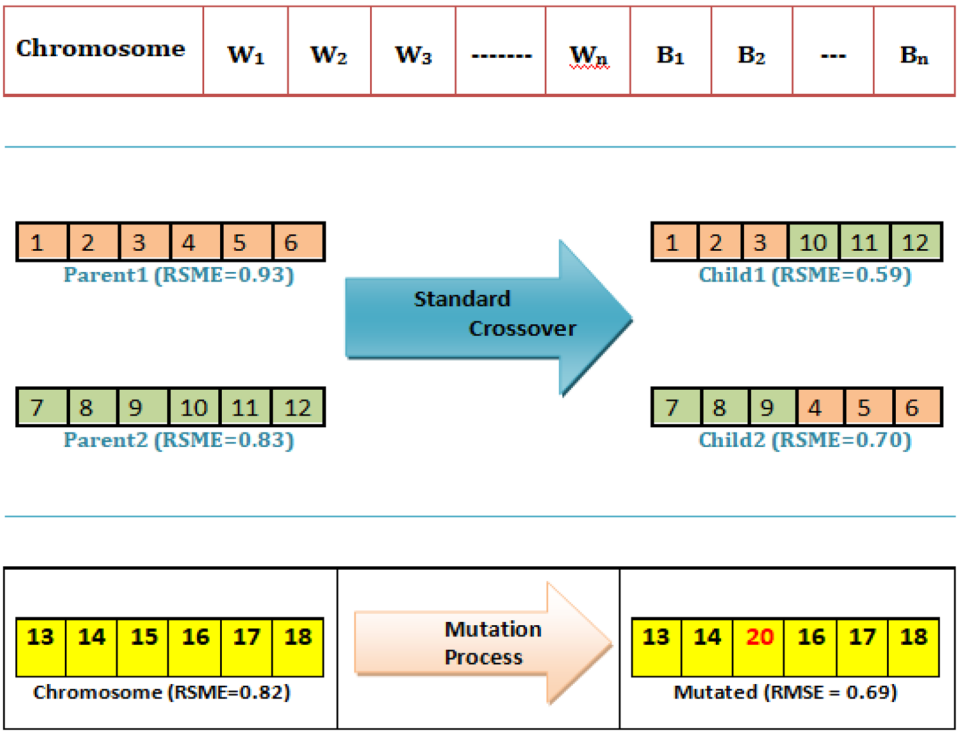


Fig. 13 Crossover and mutation operation of genetic algorithm

3.1 Genetic Algorithm (GA)

Holland invented genetic algorithm in 1975 [9], inspired by Darwin's notion of natural evolution. This algorithm mimics natural selection, where the best parents are chosen. This algorithm shuffles individual solutions. GA randomly selects current population members as parents for each generation. Over succeeding generations, the population "evolves" toward an optimal solution. Prospective readers can study articles in literature [36, 66, 70, 94, 100, 115, 117–120, 126, 132, 150–152, 164, 169, 173, 180, 184] to learn about the extensive application of GA in financial time series prediction. Several improved versions of GA, like multi-objective GA (MOGA) [73] and hybrid GA (HGA) [87], have also been suggested and shown to be good at predicting financial time series. GA comprises four phases.

1. Initial Population: A population of chromosomes starts this process. Each chromosome solves a difficulty. Genes define chromosomes.
2. Selection: Based on their fitness levels, two sets of chromosomes (parents) are chosen. Fitter chromosomes are more likely to be passed on to future generations.
3. Crossover: In a GA algorithm, this operator is the most crucial phase. A crossover point is chosen at random from the genes for each mating pair. The creation of a new organism depends on the transfer of DNA from one set of parents to another. The crossover operator is used to enhance algorithmic exploitation. This probe actually probes the area around a chromosome.
4. Mutation: Some of the genes in a newly produced offspring can be mutated. Improve exploration with the help of the mutation operator.

One of the key steps in GA modelling is defining a solution in the form of a chromosome. A chromosome, as defined in GA, a single-point crossover operation, and GA's mutation procedure is shown below in Fig. 13.

3.2 Particle Swarm Optimization (PSO)

One of the most influential intelligent optimization techniques in swarm intelligence is the PSO algorithm. The PSO technique was proposed and initially developed by Russell C. Eberhart and James Kennedy in the year 1995 [10] in order to imitate animal social interactions (e.g., a flock of birds searching for food). This algorithm mimics how members communicate and exchange information. Numerous optimisation problems have been solved using particle swarm optimization, both alone and in conjunction with other current techniques. Through agents, sometimes known as particles, whose trajectories are modified by a stochastic and a deterministic component, this technique searches for the best possible solution. The group's "best" position and each particle "best" attained position impact it, yet particles move haphazardly overall. For application of PSO in financial time series prediction interested readers

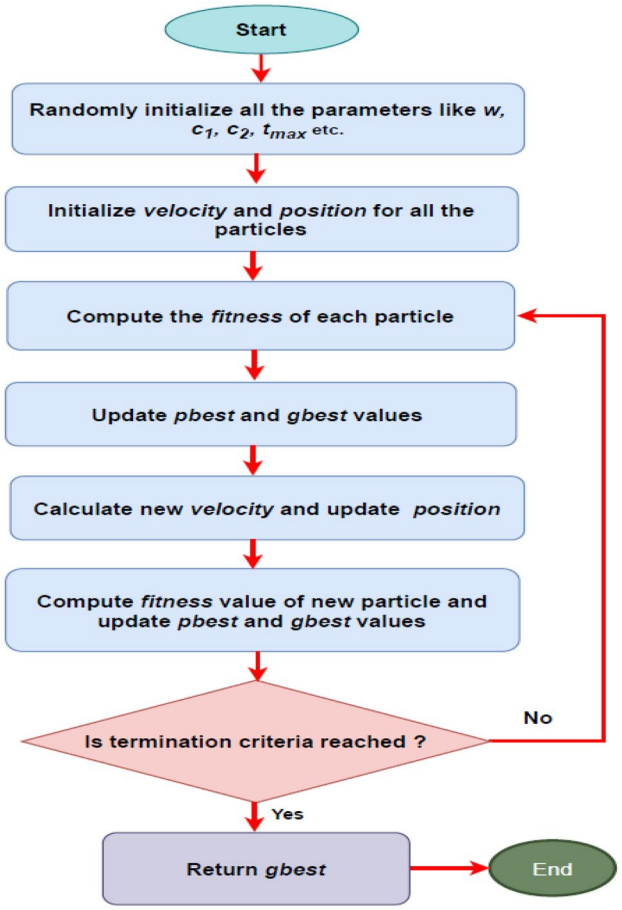


Fig. 14 Flowchart of Particle swarm optimization

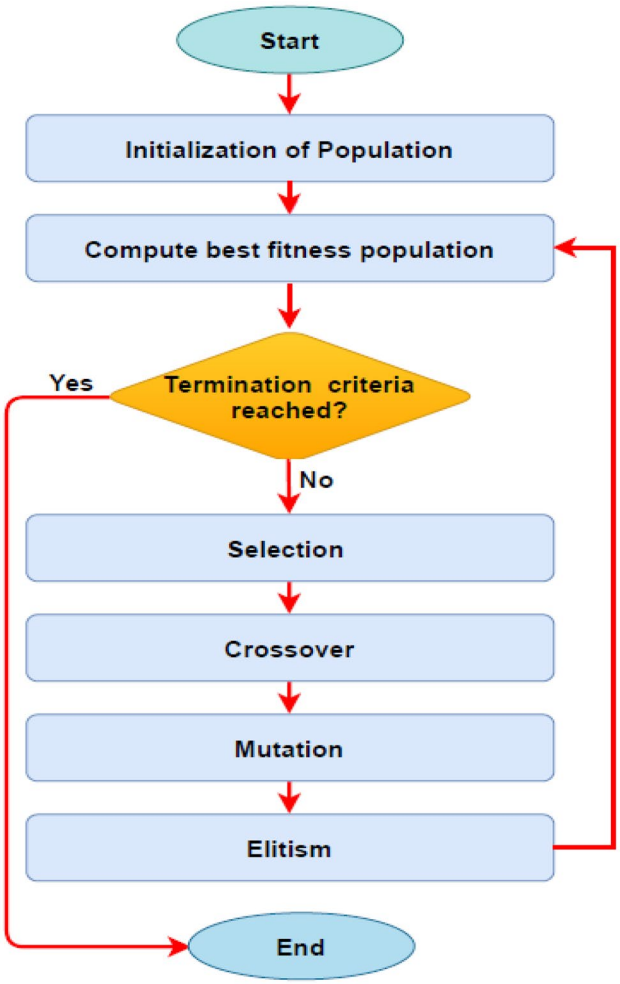


Fig. 15 Flow chart of differential evolution

may follow articles in literature [70, 78, 107, 131, 134, 138, 149, 157, 162, 165, 184–186]. To enhance the performance of PSO, some modified versions of PSO like MPSO [81, 92, 93, 97], HMRPSO [141], RWPSO & WLDPSO [146], CRPSO [137], and IPSO [76, 108, 187] are also proposed by authors. Standard working principle of PSO is given in the Fig. 14.

3.3 Differential Evolution (DE)

DE was proposed by Storn and Price in 1995 [188]. DE uses straightforward formulas to combine current candidate solutions to create new candidate solutions, maintain a population of candidate solutions, and keep the candidate solution with the highest score or fitness on the optimization task at hand. In this method, the gradient is not required since the optimization issue is viewed as a "black box" that only offers a measure of quality given a candidate solution.

Various applications of DE in financial time series prediction are given in the literature [37, 74, 85, 88, 91, 109, 123, 140, 167, 178]. The performance of a hybrid DE learning method known as "shuffled DE" (SDE) can be seen in [68].

Figure 15's flowchart explains the DE algorithm's fundamental operating concept.

3.4 Teaching–Learning Based Optimization (TLBO)

Rao et al. [15] introduced a metaheuristics technique called teaching–learning-based optimization (TLBO). The TLBO algorithm is a population-based method to problem solving that replicates the teaching and learning processes in a classroom. TLBO is divided into two phases: the instructor phase and the learner phase. The learner with the highest grade in the population is chosen to be the teacher during the teacher phase. The instructor is in charge of instructing the students and raising the class's average grade. During the learner phase, each learner chooses a learner at random to interact with. At this stage, the ultimate aim is to raise the class's mean grade. The above operations are iterated until the halting condition is reached.

The interested reader might read Zou et al., [189] for a summary of the most current TLBO improvements.

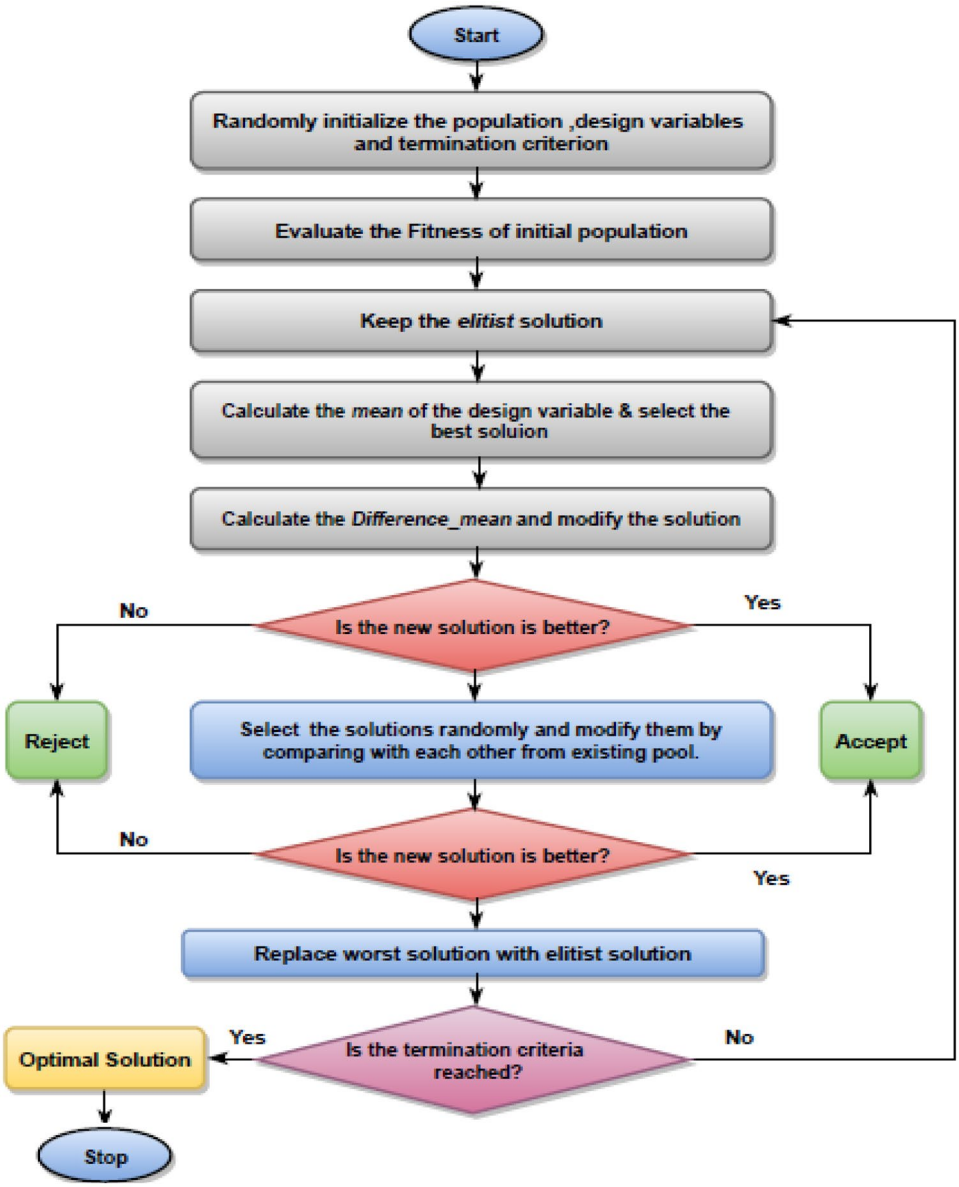


Fig. 16 The working principle of Teaching Learning Based Optimization

Literature [103] and [14] shows the contribution of TLBO in financial time series prediction. The working principle of TLBO is shown in the Fig. 16.

3.5 Chemical Reaction Optimization (CRO)

Alatas [26] in 2011 developed CRO algorithm which is persuaded by the guidelines of compound responses. Chemical reaction normally transforms substances from an unsteady state to a steady state. It finds molecules in an unstable state but excessive energy and starts with them. In a chemical system, the total amount of energy is called an enthalpy. All the way through a series of elementary reactions, molecules interact with each other and produce some intermediary chemical products. Finally, they are improved to a state with the least amount of enthalpy to support their existence. Numerous applications of CRO and its variants in financial time series prediction are seen in the literature [38, 70, 72, 77, 90, 113, 114, 124, 125, 127, 171, 177, 181, 190, 191], and the performance of a hybrid learning system, i.e., CRO-LMS and CRO-FCM, can be found in [55] and [192], respectively. The flowchart of CRO given in Fig. 17 describes the steps involved in its procedure. The optimization of CRO is given in Eq. 13 as follows:

$$MinimizeE(n) = \frac{1}{2}\sum_{k-1}^{n}\left(t_k - y_k\right)^2 \tag{13}$$

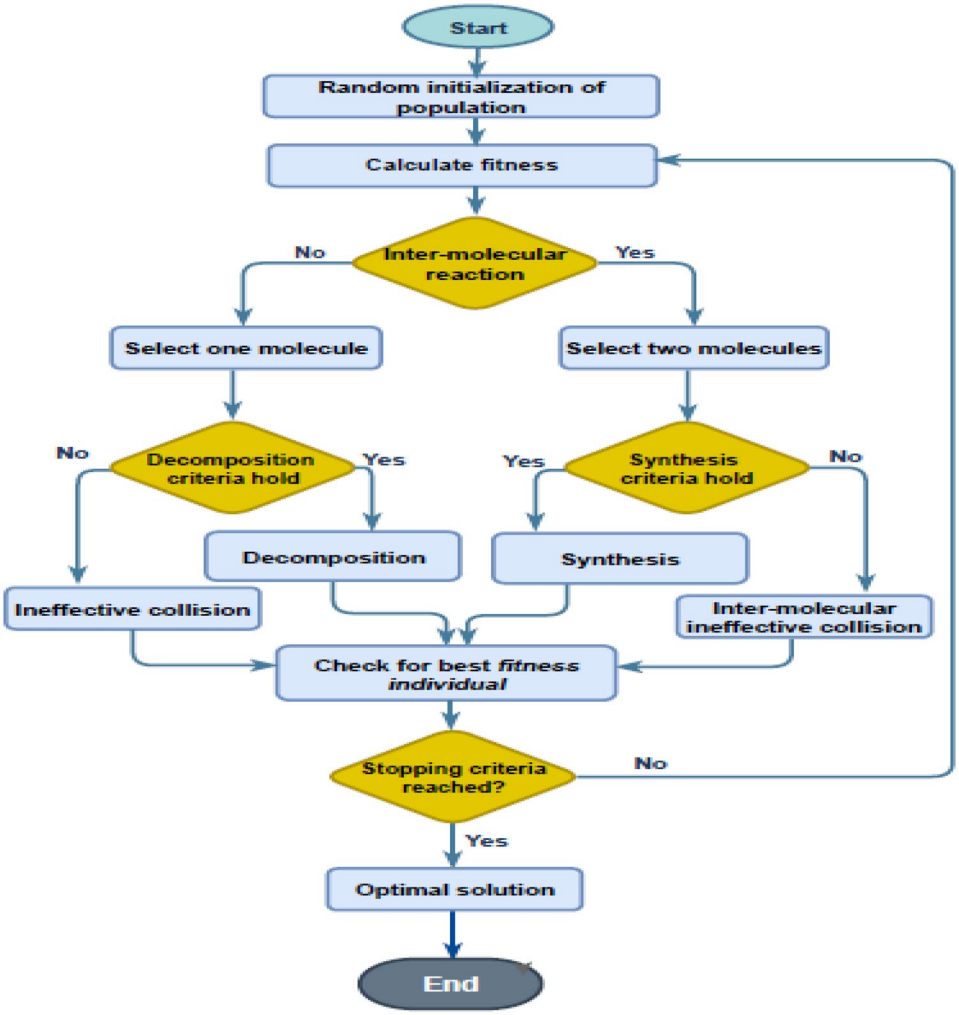


Fig. 17 Flowchart of chemical reaction optimization

here $E(n)$ is the error function, 't_k' is the actual output and 'y_k' is the estimated result.

3.6 Black Hole Optimization (BHO)

BHO was introduced by A. Hatamlou in 2013 [27]. It follows the guideline of black hole opening peculiarity. A black hole is an enormous star having huge size breakdowns. As the size of the black hole is immense and it has areas of strength for a power, so any little molecule goes over the limit of the black hole will be swallow in to the opening because of the great gravitational force of the black hole. Black hole's span is known as the Schwarzschild sweep. The significant property of such range is the speed of getting away from which is equivalent to the speed of light. The Schwarzschild sweep can be composed as follows: assessed by utilizing the condition given in Eq. 14

$$R_{BH} = \frac{2GM_{BH}}{C^2} \tag{14}$$

where R_{BH} is the Schwarzschild range, G shows for gravitational steady, M_{BH} is the mass of the dark opening; c addresses the speed of light. In our literature, article [99] addresses the application of BHO for efficient financial time series predictions. The flow chart given in Fig. 18 represents the working principle of BHO.

3.7 JAYA Algorithm

The JAYA algorithm, a contemporary metaheuristic-based algorithm, combines the characteristics of evolutionary algorithms in terms of survivability for the fittest principle, as well as Swarm Intelligence, in which the Swarm generally follows the leader during the search for optimal solution. R. Rao [193] presented the JAYA algorithm in 2016 and it has piqued the interest of a wide range of research communities

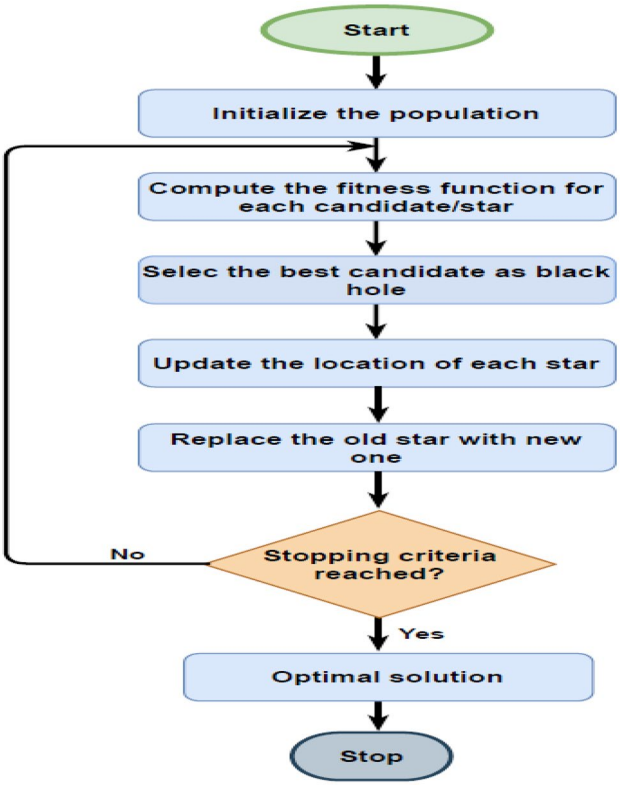


Fig. 18 Flowchart of Black hole optimization

due to its amazing characteristics: Its ideas are basic, and it is simple to use. The original search yielded no derived information. It is a parameter less algorithm. It is versatile, adaptable, and comprehensive.

For each population size $k = \{1, 2, 3, ..., n\}$ and for each iteration i, if n indicate quantity of solutions and m represents the quantity of design variables, where each chromosome is represented by each design variable, then the modified chromosome value given in the Eq. 15 as follows:

$$C^1_{j,k,i} = C_{j,k,i} + q1_{j,i}(C_{j,best,i} - \left|C_{j,k,i}\right|) - q2_{j,i}(C_{j,worst,i} - \left|C_{j,k,i}\right|) \tag{15}$$

here, C_{best} and C_{worst} are the best value and worst value obtained by the candidate respectively. But $q^1_{j,i}, q^2_{j,i}$ are the arbitrary values from $\{0, 1\}$. Figure 19 describe the working principle of JAYA algorithm. Implementation of JAYA algorithm for financial time series prediction is found in literature [35].

3.8 Grey Wolf Optimization (GWO)

S. Mirjalili et al. first put up the idea for GWO in 2013 [24]. This metaheuristic is a new optimization technology, this algorithm is completely based on hunting practice of grey wolves. Here x_1 represents the fittest solution (best wolf), the second best is x_2, and the third finest is x_3, whereas ω represents the solution for the rest of the candidates. The flow chart of GWO algorithm is given in Fig. 20 as follows.

The following are the main phases of grey wolf hunting:

1. Tracking, pursuing, and approaching the target.
2. Pursuing, surrounding, and tormenting the prey until it comes to a halt.
3. Launch an attack on the prey.

To create GWO, the social order and hunting behaviour of grey wolves are statistically represented.

3.9 Fireworks Algorithm (FWA)

The fireworks algorithm (FWA) proposed by Tan and Zhu, in 2010 [22] is a recently suggested swarm intelligence-based evolutionary algorithm. First, as fireworks, a certain number of places in the search space are picked. The fireworks then produce a series of sparks as a result of the explosion process. Following that, the superior solutions from the entire fireworks and sparks are chosen as the fireworks for the following generation to iteratively increase the quality of the solution. It has piqued the curiosity of researchers and showed good performance in many real-world issues by leveraging FWA's robust global search and information utilisation capabilities. In local space, an explosion can be visualised as a search process. If k_i is a firework and the M_i is the amplitude of an explosion, then the number of spacks C_i can be given as follows:

$$M_i = \bar{M} . \frac{f(k_i) - f_{\min} + \varepsilon}{\sum_{j=1}^{p} (f(k_j) - f_{\min} + \varepsilon} \tag{16}$$

$$C_i = \frac{m.f_{\max} - f(k_i) + \varepsilon}{\sum_{j=1}^{p} (f(k_{\max} - f(k_i) + \varepsilon} \tag{17}$$

In Eq. (13), $\bar{M}$ indicates the maximum amplitude of the explosion, $f_{\min}$, $f_{\max}$ are the minimum and maximum objective functions respectively between p fireworks, m is the controlling parameters of sparks generate by a firework and ε is a constant which is used to avoid zero division. To overcome the devastating effects of marvellous fireworks, upper and lower bounds are imposed on C_i as follows:

$$C_i = \{C_{\min}, if\ \mathrm{c_i} < c_{\min} / \mathrm{C_{max}}, if\ \mathrm{c_i} < c_{\max} / \mathrm{c_i}\ , \text{otherwise} \tag{18}$$

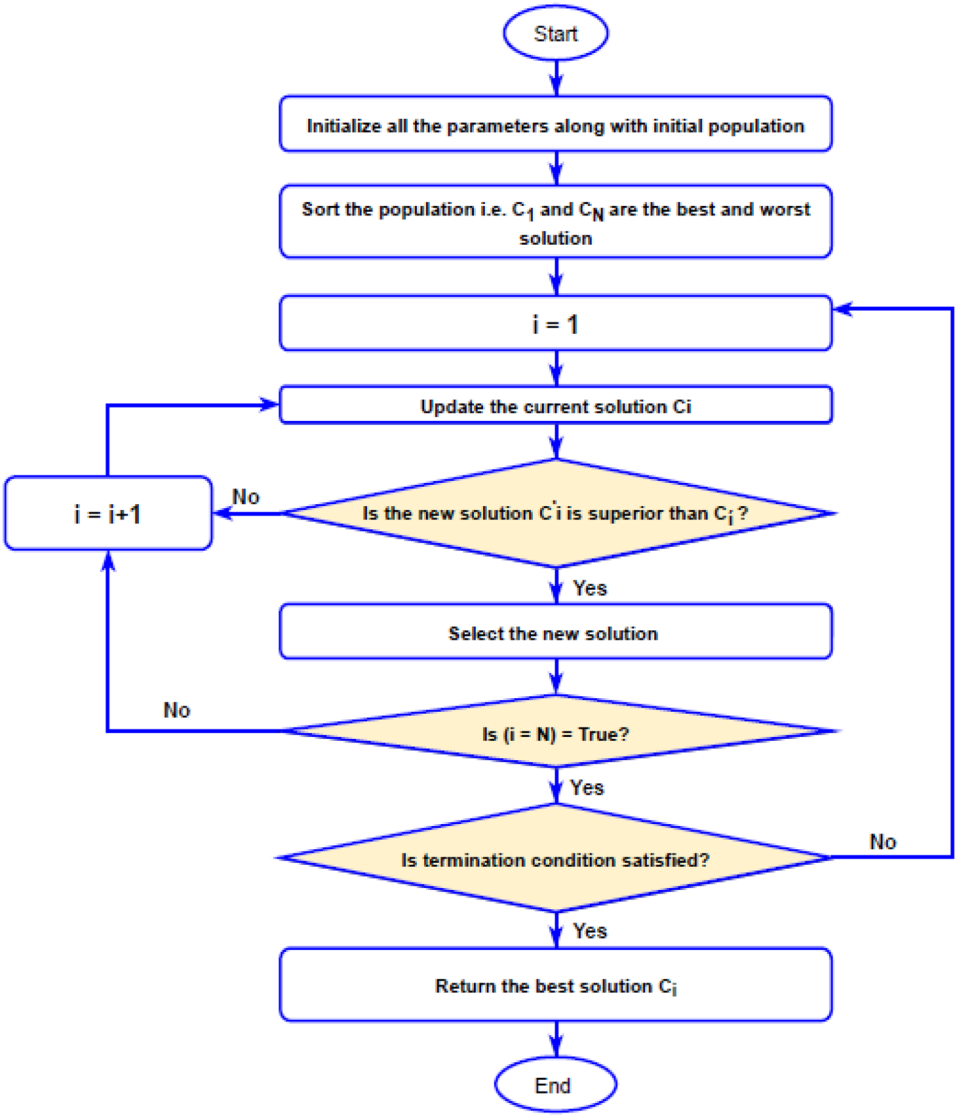


Fig. 19 The working principle of JAYA Algorithm

The location of each spark k_i generated by k_i is intended by setting z directions arbitrarily and for each dimension 't' setting the component k_j^t. Where $1 <= j <= s_i$ and $1 <= t <= z$.

The setting of k_i^t can be calculated in two ways as follows.

For most sparks, a displacement is added to k_i^t as follows:

$$K_j^t = K_i^t + A_i.rand(-1, 1) \tag{19}$$

For keeping up with variety for few explicit sparkles, a blast coefficient in view of Gaussian appropriation is applied to k_i^t as follows:

$$K_j^t = K_i^t.Gaussian(1, 1) \tag{20}$$

Another area is planned to the potential space when drops out of the pursuit space as:

$$K_j^t = K_{\min}^t + \left|K_j^t\right| \%(K_{\max}^t - K_{\min}^t) \tag{21}$$

The distance between a location K_i and another location (L) can be calculated as the sum of Euclidean distance between them as follows:

$$\text{Distance}\,(k_i) = \sum\nolimits_{j \in L} \left\| k_i - k_j \right\| \tag{22}$$

An area x_i is chosen for the cutting edge in light of likelihood esteem as follows:

$$\text{Prob}(k_i) = \frac{Dis\tan ce(k_i)}{\sum_{j \in L} Dis\tan ce(k_j)} \tag{23}$$

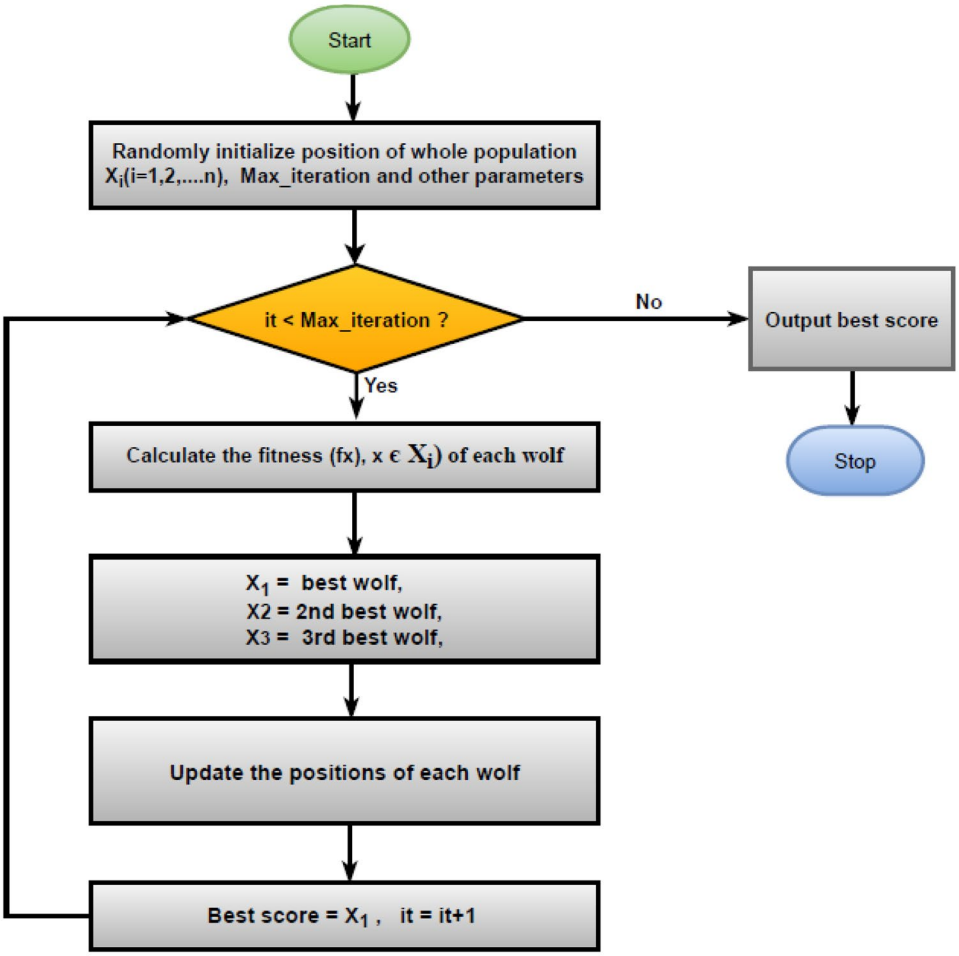


Fig. 20 Flowchart of Grey Wolf Optimization algorithm

The working principle of Fireworks algorithm is described by a flowchart given in Fig. 21. Significant FWA applications and effective FWA/HONN hybridization for time series prediction may be discovered in the literature [33, 34].

4 Hybridization of EOAs and HONNs and Financial Time Series Forecasting

To increase the prediction accuracy of ANNs, especially HONNs, EOAs are heavily used to regulate their parameters. It was discovered that they were superior to gradient descent learning in several areas, including their capacity for global minima, quick convergence, and flexibility.

A genetic algorithm (GA) was used by Nayak et al. [66] to find the best PSNN parameters for predicting stock index. Dash et al. [32] used a crow search algorithm (CSA) to train a PSNN to predict the gold/INR and gold/AED exchange rates. When CSA was compared to the PSO and DE algorithms, CSA did a better job of training the model than the other two. Nayak [33] used the Fireworks algorithm (FWA) to optimize the PSNN's parameters when trying to predict the price of crude oil. He found that the FWA-based PSNN got to global minima more quickly. Sahu et al. [55] found that when training different kinds of FLANN to predict foreign exchange rates, PFLANN and LFLANN trained by CRO-LMS did better than LMS-FLANN and CRO-FLANN in most cases. This is because the hybrid learning's faster convergence and search refinement features made the models more accurate. Nayak [67] used GA and PSO, two evolutionary optimization techniques, to optimize the parameters of four HONNs (PSNN, SPNN, JPSNN, and FLANN) in order to create a number of hybrid models for predicting currency exchange rates. He found that the PSO-HONN and GA-HONN hybrid models did better than other traditional models.

Das et al. [102] made Rao algorithm–based FLANNs (RAFLANNs) hybrid models to predict stock closing prices accurately so they could study how well the algorithm works. Nayak et al. [35] used the JAYA algorithm to find PSNN's best parameters so that they could make a hybrid model called JPSNN. This model was used to predict stock indices. Dash et al. [68] used a hybrid learning method

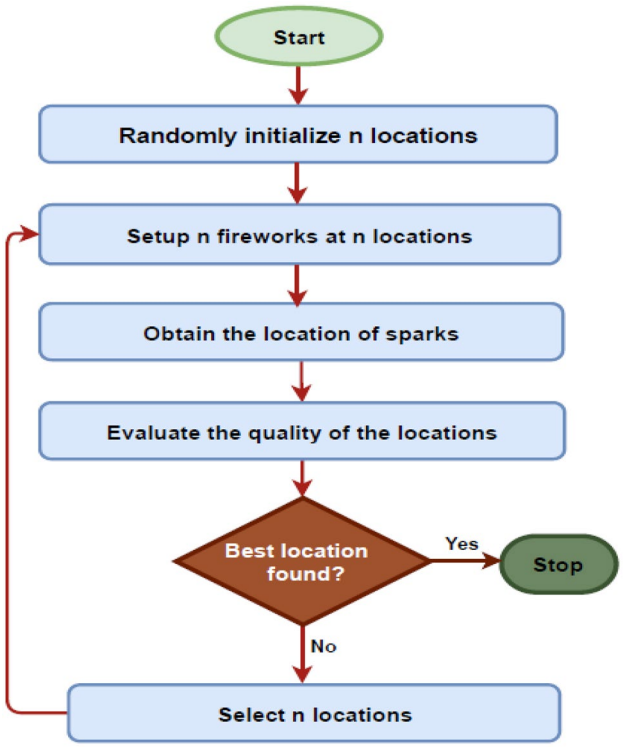


Fig. 21 The working principle of Fireworks algorithms

known as the Shuffled Differential Evolution algorithm to develop a Pi-Sigma Neural Network-based Predictor Model called PSNN-SDE, which they then applied to the problem of predicting the USD/CHF and USD/JPY exchange rate prices. Nayak et al. [21] made a hybrid classifier called FFA-PSNN that is based on the firefly PSNN. Ylmaz et al. [37] used a differential evolution algorithm and a pi-sigma artificial neural network to create a hybrid model that can predict financial time series for AUST and TEC. To accurately predict the fuzzy time series, Patnaik et al. [74] designed a hybrid model called DE-PSNN, where DE is used to update the optimum parameters of PSNN.

Using honey bee mating optimization (HBMO) to determine PSNN's best settings, Nayak and Naik [75] created a hybrid model they called HBMO-PSNN to use for classification. Kanungo et al. [76] suggested a better PSO-based PSNN classifier (IPSO-PSNN) for financial data. R. Dash [71] optimized PSNN parameters using the ISFL algorithm to anticipate USD/CAD, USD/CHF, and USD/JPY exchange rates. Patnaik et al. [72] predict fuzzy time series using CRO-PSNN. Sahoo et al. [73] used a PSNN-MOGA hybrid model to predict BSE stock prices. Nayak et al. [14] created a time series data PSNN-TLBO classifier. Nayak et al. [94] trained an adaptable evolutionary PSNN (AE-PSNN) using PSO and GA and compared its performance to GD learning. PSO and GA learning beat GD learning. Nayak et al. [78] used PSO to train a PSNN for excellent classification accuracy. Nayak et al. [77] suggested a CRO-based PSNN (CRO-PSNN) as a hybrid classifier for nonlinear classification and compared its performance with GA-PSNN, PSO-PSNN, and PSNN. The proposed technique is quick, stable, and dependable and delivers superior classification accuracy. Chen et al. [178] used three algorithms—ACO, ABC, and DE—to find the ideal hybrid neural network (HNN) parameters for downstream river flow forecasting. They discovered that the DE and ACO strategies are best for finding HONN model parameters such as the recession coefficient and initial storage.

Cao [179] said that feed-forward neural networks (FNNs) could be trained using interactive phase-based optimization (IPBO). Shrivastava and Verma [135] created a hybrid swarm optimization to train a Cascaded SOM-RBF (CSR) model for option price prediction using PGO and ACO. Bas et al. [81] used a modified PSO to find the ideal PSNN values for high-order fuzzy time series forecasting. Nayak et al. [86] suggested using a hybrid PSO-GA-based Pi Sigma Neural Network (PSNN) for classification. Nie and Deng [87] trained a PSNN using a hybrid genetic algorithm (HGA) for function optimization. The HGA outperformed the conventional GA. Dash et al. [88] used PSO and DE algorithms to create a PSNN-based gold price predictor. Sahu et al. [90] suggested CRO-FLANN with LMS for foreign currency forecasting. Yılmaz et al. [85] trained DEA-PS-ANNs for financial time series forecasting. Arsian and Yolcu [97] used a modified PSO to train a hybrid sigma-pi neural network (HSPNN) for integrated intuitiveistic fuzzy time series prediction. Akdeniz et al. [92] developed an ARMA-type pi–sigma artificial neural network (ARMATPS-ANN) for nonlinear time series forecasting using a modified PSO (MPSO) learning method. Sudeepa et al. [147] used cluster-based modified CSA optimized ELM and intrinsic time-scale decomposition to predict stock market prices (ITD-CMCSA-OELM).

Mili and Hamdi [116] suggested a hybrid FLANN (HFLANN) model for data mining and classification, employing three population-based optimization methods: GA, PSO, and DE. PSO performed well in optimizing the model. Behera et al. [113] suggested a CRO-FLN to enhance software reliability prediction. Akbilgic et al. [132] used a GA to pick the best predictors for a new Hybrid RBF Neural Networks (HRBF-NN) model that could predict the Istanbul Stock Exchange National 100 Index (ISE100) and seven other international stock market indices. Egrioglu et al. [19] came up with the idea of using pi-sigma artificial neural networks trained by artificial bee colonies (ABC) to make intuitive high-order fuzzy time series forecasts. Zhang and Hu [148] created ultra-high-frequency sigmoid and cosine artificial higher-order neural networks (UGC-HONN) for exchange rate forecasting. Das et al. [118] proposed financial time series forecasting using a GA-based polynomial

FLANN with virtual data points. Nayak et al. [194, 195] used Rao algorithm-based artificial neural networks (RA-ANN) to predict bitcoin closing prices. Worasucheep et al. [170] used a recent variation of an artificial bee colony, ABCDE, to optimize the weights of a three-layer feedforward neural network (FFNN) to anticipate USD/EUR and USD/Yen exchange rates. Waheeb and Ghazali [120] estimate daily exchange rates using a genetically optimized tensor product functional link neural network (GA-TPFLANN). Nayak and Ansari [69] suggested a cooperative optimization algorithm-based HONN (COA-HONN) for stock forecasting and compared it to GA-HONNs and CRO-HONNs. CRO-HONNs excelled. Sahu et al. [91] developed three evolving higher order neural networks (EHONN) and evaluated their performance on modeling and forecasting five exchange rate time series using a pi-sigma neural network (PSNN) as a base model. The optimal architectures of PSNN are evolved on the fly with three evolutionary learning methods: genetic algorithms (GA), differential evolution (DE), and particle swarm optimization (PSO), forming three hybrid models.

For the purpose of predicting financial time series, Dash et al. [130] suggested a self-adaptive differential harmony search-based optimized extreme learning machine (SADHS-OELM). In order to forecast the stock market, Soumya et al. [17] used a FLANN-HS, which is an artificial neural network based on the principle of harmony search. Anish and Majhi [121] came up with a nonlinear adaptive system that uses feedback fuzzy linear array neural networks (FLANNs) along with recursive least square (RLS) training and factor analysis. An adaptive bioinspired Firefly algorithm (FFA) is used to find the best weights for the proposed model, which is based on the work of Rout et al. [122] on a computationally efficient recurrent functional link neural network (RCEFLANN) for time series forecasting. Nayak et al. [125] proposed a functional link network based on CRO (artificial chemical reaction optimization) and dubbed it ACFLN for stock market prediction. For non-linear data categorization, Naik et al. [104] created a self-adaptive harmony search-based functional link higher-order artificial neural network (SAHS-FLANN). Naik et al. [105] suggested HBMO-GDL-FLANN for categorization. Zhao and Yang [137] used a PSO-based single multiplicative neuron model for time series prediction with cooperative random learning particle swarm optimization (CRPSO) to increase PSO performance.

Chouikhi et al. [157] made an ESN-PSO hybrid model for forecasting time series by using particle swarm optimization (PSO) to optimize echo state network (ESN) parameters in an efficient way. Zhong et al. [164] made a genetic algorithm (GA)-optimized double-reservoir echo state network (DRESN) for multi-regime time series prediction. Nayak et al. [124] devised an artificial chemical reaction (ACRO) of recurrent functional link networks (ACRRFLN) for stock market prediction, which outperformed two previous methods. GA and PSO also trained the model. CRO learning outperformed the other two. S.C. Nayak [126] came up with optimal functional link neural networks (FLNs) for predicting the closing price of Bitcoin and used a genetic algorithm (GA) to update the weight and bias vector of FLNs and the best number of polynomial functions for each category of basis functions. Rout et al. [123] used a recurrent computationally efficient functional link neural network (RCE-FLANN) for financial time series forecasting. They used several evolutionary learning methods, including Particle Swarm Optimization (PSO), HMRPSO, and Differential Evolution (DE), to find the optimal weights of the proposed model. DE outperformed the others. Naik et al. [106] suggested a unique FLANN for classification using hybrid PSO and GA-based gradient descent learning.

Naik et al. [103] presented opposition-based learning-TLBO (OLE-TLBO) to categorize nonlinear data. Sahu et al. [34] designed a fireworks algorithm-based functional link network (FWA-FLN) to forecast currency exchange rate time series. They focused on training data enrichment. Multi-verse optimal multiplicative functional link neural networks were used by Sahu et al. [172] to estimate south Asian currency rates multi-steps ahead (MVO-MFLN). In order to facilitate categorization, Dehuri et al. [108] proposed a swarm-optimized functional link artificial neural network (ISO-FLANN). Mahapatra et al. [109] came up with a way to predict how the Indian stock market will do based on a functional link artificial neural network (FLANN) trained through differential evolution. The functional link network with artificial chemical reaction optimization (FLN-ACRO) was used for stock market forecasting by Nayak et al. [127]. Four low-complexity, single-layer forecasting models were constructed by Nayak et al. [119], namely TFLANN, CFLANN, LeFLANN, and LFLANN; the weight and bias vectors and the number of functional expansions for each model were optimized using a genetic algorithm. Behera and Behera [196] used the firefly algorithm (FFA) to train the dataset and discovered that FFA-RPNN outperformed FFA-MLP in all datasets. Nayak et al. [176] developed an elitist artificial-electric-field-algorithm-based artificial neural network (eAEFA+ANN) for financial time series forecasting. Nayak et al. [177] suggested ANNCRO to forecast the daily closing prices of 10 fast-growing stock markets: the BSE, DJIA, NASDAQ, FTSE, TAIEX, S&P 500, ASX, LSE, SSE, and NIKKEI. Wang et al. [155] created a GWO ESN time series prediction model. Luo et al. [36] came up with a deep feed forward neural network (GA-DFNN) architecture that uses a genetic algorithm to predict how much power a real building will use. A multiobjective GA (NSGA-II)-based RBF adaptive stock index prediction model was made by Majhi et al. [129]. Dash [140] created a hybrid stock predictor utilizing Differential Evolution and Chebyshev Polynomial Neural Networks (DECPNN).

Dash and Dash [141] used moderate random search particle swarm optimization to estimate LENN model parameters to forecast financial time series data (HMRPSO). Nayak and Misra [38] developed extreme learning with chemical reaction optimization, a learning system that combines ELM and CRO (ELCRO). A single-layer feed-forward neural network's weight, bias vector, and hidden neurons are optimized via ELCRO. Based on modified PSO and GRNN, Wang et al. [146] built Taiwan dollar to US dollar and RMB exchange rate prediction models (RWPSO-GRNN and WLDPSO-GRNN). Dash [154] used a recurrent Legendre polynomial neural network (RLPNN) and an improved shuffling frog leaping (ISFL) learning method to make a hybrid FOREX prediction model (RLPNN-ISFL). Egrioglu et al. [149] used data from the 2016 international time series forecast competition (CIF-2016) to test a particle swarm optimization (PSO)-trained Median-Pi artificial neural network (MdPNN). Nayak and Misra [150] used a GA-weighted condensed polynomial neural network (CPNN-GA) to calculate stock closing indices. They tested how well it worked by predicting the closing price of five fast-growing stock indices for the next day: the BSE, DJIA, NASDAQ, FTSE, and TAIEX. Nayak et al. [151, 152] made an adaptive second-order neural network with genetic algorithm-based training (ASONN-GA) for predicting financial time series. Nayak et al. [99] made a black hole optimization and higher order Jordan Pi-sigma neural network (BHO-JPSNN) for tasks that don't follow a straight line.

Nayak et al. [100] used gradient descent learning to show how a higher-order evolutionary Jordan pi-sigma neural network (JPSNN) can be used to classify time series. Kumar [131] developed nine new integrated models for intraday stock price forecasting using three ANNs, i.e., BPNN, RBFNN, and TDNN, plus nature-inspired algorithms like GA, PSO, and Artificial Bee Colony (ABC). Models included PSO-BPNN, ABC-BPNN, GA-BPNN, GA-RBFNN, PSO-RBFNN, ABC-TDNN, and GA-TDNN. Nayak et al. [181] proposed intelligent financial forecasting with an improved chemical reaction optimization algorithm based on a dendritic neuron model (ICRO-DNM) and compared the performance of ICRO learning with CRO, PSO, and GA. They found that the combination of ICRO's powerful global search and DNM's better approximation gave them better results. Dash et al. [114] suggested and tested a CRO-FLANN classifier for a non-imputed and inconsistent dataset. Husaini et al. [79, 80] used a modified cuckoo search-Markov chain Monte Carlo (MCS-MCMC) learning technique for weight optimization in higher order neural networks (HONN) models (FLNN-MCMC, PSNN-MCMC) for predicting physical and financial time series. Aladag et al. [162] made a model for real seasonal time series forecasting that is based on particle swarm optimization (PSO) and a multiplicative seasonal neural network (MSANN).

Egrioglu et al. [165] used PSO to train a recurrent multiplicative neuron model artificial neural network (RMNM-ANN) for non-linear forecasting of time series. Dehuri and Cho [117] made the HFLNN for classifying by using a GA to combine features that have gotten better over time with a functional link neural network. With the help of a differential evolution algorithm, Bas [167] taught a multiplicative neuron mode-based artificial neural network (MNM–ANN–DEA) to predict real-world time series data. Smith and Jin [3] say that a hybrid multi-objective evolutionary algorithm (HMOEA) is used to train and improve recurrent neural networks (RNNs) used to predict time series. Kocak et al. [93] created an ARMA-type Pi-Sigma neural network (ARMATPS-ANN) and trained it by PSO for fuzzy time series forecasting. Rather et al. [169] used a genetic algorithm (GA) and a RNN to make a hybrid prediction model (HPM) to predict stock returns. Asadi et al. [180] suggested using genetic algorithms (GA) and LM algorithms for learning feed-forward neural networks to combine evolutionary Levenberg–Marquardt neural networks (PELMNN) and data pre-processing to predict the stock market.

Chang et al. [173] suggested an evolving partly connected neural network (EPCNN) to predict stock price trends and used a GA to update the model's best parameters. Yogi et al. [107] proposed a PSO-based FLANN training method for equalizing digital communication channels. Nayak and Misra [171] suggested a chemical-reaction-optimization-based neuro-fuzzy hybrid network (CRO-NFN) for predicting the closing price of a stock. Yolcu et al. [138] suggested a single multiplicative neuron model artificial neural network with an autoregressive coefficient (AC-S-MNM) and trained it using a PSO for time series modelling. Tang et al. [174] used an adaptive differential evolution algorithm with an optional external archive (JADE) to train the EPNN model for financial bankruptcy analysis. Das et al. [168] successfully predicted stock market prices using an extreme learning machine (ELM) optimized using the PSO-based group-oriented crow search (PGCSA) algorithm. Dehuri and Cho [115] came up with a hybrid genetic-based functional link artificial neural network (HFLANN) classifier. T. Xiong [134] improved the model's structure and parameters for interval time series forecasting by using a fully complex-valued RBF neural network (FCRRBFNN) with DPSO and PSO algorithms. H. Emam [197] created stock exchange trading optimization (SETO) to address numerical and technical issues. Yang and Duan [175] came up with an effective co-evolutionary algorithm based on an artificial bee colony and differential evolution (called "improved ABC–DE"), and they taught an extreme learning machine (ELM) to predict time series.

Table 2 Efficacy of EOA-based HONN used for financial time series forecasting

Proposed EOA	Hybridization of EOA and HONN	Types of parameters optimized	Observations
GA [66]	GA-PSNN	Optimal parameters	Superior to gradient descent algorithm in optimizing the weights and biases
CRO-LMS [55]	CRO-FLANN	Optimal parameters	CRO provides faster convergence, LMS provides the search refinement
GA and PSO [67]	GA-HONN, PSO-HONN	Optimal parameters	Variants of HONN are successfully trained by GA and PSO
RA [102]	RAFLANN	Optimal parameters	Outperformed to MVO, BMO, GA, and GD learning
CSA [32]	CSA-PSNN	Optimal parameters	Superior to PSO and DE learning
OLE-TLBO [103]	OLE-TLBO-FLANN	Optimal parameters	Outperformed to GA, PSO, TLBO, E-TLBO, and OL-TLBO learning
FWA [33]	FWA-PSNN	Optimal parameters	Capability of faster convergence and global optima achieved
FWA [34]	FWA-FLN	Optimal parameters	FWA learning provides better prediction accuracy than DE, GA, and PSO
ELCRO [38]	SLFN	Optimal & hyper parameters	Faster convergence, better results than GA, PSO and GD learning
JAYA [35]	JAYA-PSNN	Optimal parameters	Better learning achieved than GD, GA and PSO
SDE [68]	PSNN-SDE	Optimal parameters	PSNN-SDE outperform DE, PSO and SFL learning
GA [118]	FLANN	Optimal parameters	Better input–output mapping and high accuracy achieved
COA [69]	COA-PSNN, COA-FLANN	Optimal parameters	Better optimization of optimal parameters than GA and CRO and enhances accuracy
GA, DE and PSO [91]	GA-PSNN, PSO-PSNN, DE-PSNN	Optimal parameters	PSO based model perform relatively better than others
CRO [125]	ACFLN	Optimal parameters	Faster convergence and better accuracy
DEA [37]	DEA-PS-ANN	Optimal parameters	DE has less parameters, global optima and better results achieved
RWPSO and WLDPSO [146]	RWPSO-GRNN, WLDPSO-GRNN	Optimal parameters	RWPSO-GRNN is found superior to WLDPSO-GRNN
FFA [21]	FFA-PSNN	Optimal parameters	Steady, reliable prediction and better accuracy, outperformed GA, PSO and PSO-GA learning
GA [36]	GA-DFNN	Optimal and hyper parameters	GA determines both the optimal and hyper parameters of the network and in result better prediction accuracy and less computation complexity achieved
DE [140]	DECPNN	Optimal parameters	DE produces better APE with faster convergence rate than both BP and PSO learning
HMRPSO [141]	LENN	Optimal parameters	HMRPSO perform random search, enhances the global search ability, and achieved faster convergence. Outperforms PSO and DE learning
GA [120]	GA–TPFLNN	Optimal parameters	GA helps the TPFLNN to find low complexity network structure and/or near-optimum parameters
HS [17]	FLANN-HS	Optimal parameters	Faster convergence, harmony search outperforms DE and BP learning
GA [180]	PELMNN	Optimal & hyper parameters	Good prediction accuracy found and GA is suitable for input–output mapping

Table 2 (continued)

Proposed EOA	Hybridization of EOA and HONN	Types of parameters optimized	Observations
ACO, ABC &DE [178]	Hybrid- NN	Optimal parameters	DE found to be superior and provides higher accuracy, ABC provides slow convergence, and DE is more adaptive and provides good optimization
HMOEA [3]	RNN	Optimal & hyper parameters	HMOEA uses a GA plus local search, good structure optimization
FFA [122]	RCE-FLANN	Optimal parameters	FFA is found to be adaptive, and increases the performance of the network
MPSO [92]	ARMATPS-ANN	Optimal parameters	Better prediction results achieved faster convergence
GA [169]	HPM	Optimal parameters	GA is used to find the optimal weights of HPM and excellent prediction results achieved
PSO & GA [88]	PSNN	Optimal & hyper parameters	Better results achieved by DE learning than PSO learning
ABCDE [170]	FFN	Optimal parameters	ABCDE found superior to ABC, BP & ODE in training the models optimal parameters
DPSO/ PSO [134]	DPSO/ PSO -FCRBFNN	Optimal & hyper parameters	DPSO / PSO is used to jointly optimize the structure and parameters of the model and produce efficient results
RLS [121]	FFLANN	Optimal parameters	RLS reduces training time, requires less iterations as compared to others
ISFL [71]	PSNN	Optimal parameters	ISFL found superior in training the network than other learning algorithms
CMCSA [147]	ITD-CMCSA-OELM	Optimal parameters	The hybrid ITD-CMCSA outperforms CMCSA, CSA, DE and other learning
MOGA [73]	PSNN-MOGA	Optimal parameters	Global optima achieved with faster convergence rate, MOGA outperforms GA and DE
PSO, HMRPSO & DE [123]	RCE-FLANN	Optimal parameters	DE/current to best mutation provides good result than PSO and HMRPSO
PGCSA [168]	PGCSA ELM	Optimal parameters	The effectiveness of conventional ELM has improved by PGCSA
UGC [148]	UGC-HONN	Optimal parameters	HONNs trained by UGC learning are 2.8546% to 3.4185% better than others
PSO [149]	MdPNN	Optimal parameters	MdPNN performed well when trained by PSO than BP learning
SETO [197]	–	–	Faster convergence, efficient exploitation, exploration, and local avoidance ability of SETO found better than others
CRO [171]	CRO-NFN	Optimal parameters	The optimal search space has explored by CRO, which requires fewer tunable parameters and possess faster convergence
GA [150]	CPNN-GA	Optimal parameters	GA efficiently determined the parameters of CPNN
GA [126]	FLN Variants	Optimal & hyper parameters	Better forecast accuracies achieved, but use of GA may add computational cost due to genetic operators
GA [152]	ASONN-GA	Optimal parameters	Due to global search and good parameter tuning, higher accuracy found
PSO & GA [94]	AE-PSNN	Optimal parameters	Both PSO and GA are found superior to GD learning

Table 2 (continued)

Proposed EOA	Hybridization of EOA and HONN	Types of parameters optimized	Observations
PSO, GA and ACRO [124]	ACRRFLN	Optimal parameters	ACRO has a smaller number of tuning parameters, faster convergence and found capable to capture the uncertainties very well than others
BHO [99]	BHO-JPSNN	Optimal parameters	BHO is superior to GA, PSO and IPSO in training the network
DE [74]	DE-PSNN	Optimal parameters	DE is more effective due to less parameter, DE outperformed to Jaya and CRO
HBMO [75]	HBMO-PSNN	Optimal parameters	HBMO is reliable and provides better accuracy and outperformed to PSO & DE
IPSO [76]	IPSO-PSNN	Optimal parameters	IPSO outperformed to PSO & GA in terms of accuracy
SAHS [104]	SASH-FLANN	Optimal parameters	SASH is significant in training the network than HS & IHS learning
HBMO-GDL [105]	HBMO-GDL-FLANN	Optimal parameters	HBMO free from mutation & crossover and relatively better than GA and PSO
TLBO [14]	PSNN-TLBO	Optimal parameters	TLBO found superior to others
GA [100]	GA-JPSNN	Optimal parameters	Average classification accuracy, and better results found
GA [151]	ASONN-GA	Optimal parameters	(Global search ability of GA) + (generalization ability of ASONN) = handling the uncertainties and nonlinearities associated with the financial time series
CRO [77]	CRO-PSNN	Optimal parameters	Global minima achieved due to good optimization of optimal parameters
PSO–GA [106]	PSO-GA-FLANN	Optimal parameters	Hybrid learning perform relatively better than individual GA & PSO learning
GA [119]	FLANN variants	Optimal & hyper parameters	GA determined the functional expansion of input signals and other optimal parameters very well
NSGA-II [129]	MORBF	Optimal & hyper parameters	NSGA-II found to be superior to other in training algorithm
SADHS-OELM [130]	RBF, CEFLANN	Optimal & hyper parameters	The hybrid learning SADHS-OELM outperforms other learning methods and CEFLANN outperformed for both stock index and volatility prediction
PSO [78]	PSNN	Optimal parameters	PSO found to be robust algorithm with global search capabilities than GA
GA [173]	EPCNN	Optimal parameters	Weights are optimized by GA and EPCNN is found to be convergent
MCS-MCMC [79]	FLNN-MCMC, PSNN-MCMC	Optimal parameters	The MCS-MCMC hybrid learning helped in dealing with slow convergence and poor generalization
MCS-MCMC [80]	FLNN-MCMC, PSNN-MCMC	Optimal parameters	MCS modifies CS parameters to increase convergence, while MCMC simplifies objective function generation
JADE [174]	EPNN	Optimal & hyper parameters	Global optimization convergence beats MLP and PNN
ISFL [154]	RLPNN-ISFL	Optimal & hyper parameters	Good nonlinear mapping seen between input–output; excellent outcomes were obtained

Table 2 (continued)

Proposed EOA	Hybridization of EOA and HONN	Types of parameters optimized	Observations
MPSO [81]	PSNN	Optimal parameters	The proposed learning method trained the models parameters to improve accuracy
GA, ABC, and PSO [131]	{GA, PSO, ABC}-BPNN, {GA, PSO, ABC}-TDNN, {GA,PSO,ABC}-RBFN	Optimal parameters	The PSO-BPNN model yielded the highest prediction accuracy
Improved ABC-DE [175]	Improved ABC-DE-ELM	Optimal parameters	Hybrid learning improves generalisation, training time, parameter optimization, and accuracy
ACO &PGO [135]	Cascaded SOM-RBF (CSR)	Optimal parameters	Hybrid learning reduces the variation of attributes and produces efficient results
GWO [155]	GWO_ESN	Optimal & hyper parameters	In terms of accuracy, GWO_ESN > PSO_ESN > RSN model > Elman model > BP model
DE [85]	DEA-PS-ANN	Optimal parameters	PSANN trained by DEA outperformed
PSO-GA [86]	PSO-GA-PSNN	Optimal parameters	The hybrid learning provides better accuracy than PSO, GA and other
GA [132]	HRBF-NN	Optimal & hyper parameters	Indices which impact ISE100 index were properly determined
HGA [87]	PSNN	Optimal & hyper parameters	Better convergence achieved
GA [117]	HFLNN	Optimal parameters	Optimal parameters of HFLNN optimized by GA and better results achieved
CRO [90]	CRO-FLANN with LMS	Optimal parameters	CRO based FLANN Model trained with LMS performs better
PSO [157]	ESN-PSO	Optimal & hyper parameters	An improvement in generalization and faster convergence achieved
GA [164]	DRESN	Optimal & hyper parameters	GA is effective in solving mixed-integer problem
PSO [107]	FLANN	Optimal parameters	PSO algorithm performed far more efficiently than other two
ISO (IPSO) [108]	ISO-FLANN	Optimal parameters	The performance of FLANN trained by ISO for classification task is promising
DE [109]	FLANN	Optimal parameters	The DE optimized FLANN is proving it's superiority
MPSO [97]	HSPNN	Optimal parameters	MPSO determined optimal weights and excellent results achieved
PSO [162]	MSAAN	Optimal parameters	The weights and biases of the network fine tuned to obtained outstanding results
PSO [165]	RMNM-ANN	Optimal parameters	PSO successfully trained the model in order to achieve good results
DE [167]	MNM-ANN-DEA	Optimal parameters	The model has trained by DEA successfully
PSO [138]	AC-S-MNM-PSO	Optimal parameters	PSO successfully determined the coefficients of autoregressive equations, initial values of the weights and biases
CRO [72]	CRO-PSNN	Optimal parameters	CRO learning outperformed TLBO and JAYA learning
CRPSO [137]	CRPSO-SMNM	Optimal parameters	The model is found to be efficient and robust in case of CRPSO learning rather than GA, PSO and BP learning
GA [115]	HFLANN	Optimal parameters	Weights and biases of HFLANN optimized by GA and better results achieved

Table 2 (continued)

Proposed EOA	Hybridization of EOA and HONN	Types of parameters optimized	Observations
ABC [19]	PS-ANN	Optimal parameters	ABC is a strong numerical optimization algorithm and hence employed to determine the parameters of the PS-ANN model
GA, DE, and PSO [116]	HFLANN	Optimal parameters	PSO is the best local optima and DE is the best global optima
MPSO [93]	ARMATPS-ANN	Optimal parameters	MPSO determined the parameters of the model efficiently
CRO [177]	ANNCRO	Optimal parameters	CRO has less parameter and provides faster convergence as well as better approximation
ACRO [127]	FLN-ACRO	Optimal parameters	ACRO based model produces better forecasting results
CRO [113]	CRO-FLN	Optimal parameters	CRO performed global search and able to overcome the well-known pitfalls of the GD-based learning
MVO [172]	MV-MFLN	Optimal parameters	MVO outperformed to GA, DE, TLBO, and GD learning
ICRO [181]	ICRO-DNM	Optimal & hyper parameters	The powerful global search ability of ICRO synergies with better approximation capability of DNM. ICRO learning outperformed to CRO, PSO and GA in terms of faster convergence and accuracy
CRO [114]	CRO-FLANN	Optimal parameters	Faster convergence and better accuracy achieved
Elitism AEFA [176]	eAEFA+ANN	Optimal parameters	The optimal neuron parameters of eAEFA+ANN are decided by eAEFA, better performance seen
IPBO [179]	FNN	Optimal parameters	IPBO has ability to balance local & global search and outperforms other
RA [194]	RA-ANN	Optimal parameters	Faster convergence & better generalization ability of RA has been seen
FFA [196]	FFA-RPNN	Optimal parameters	FFA has shown better result in all different datasets

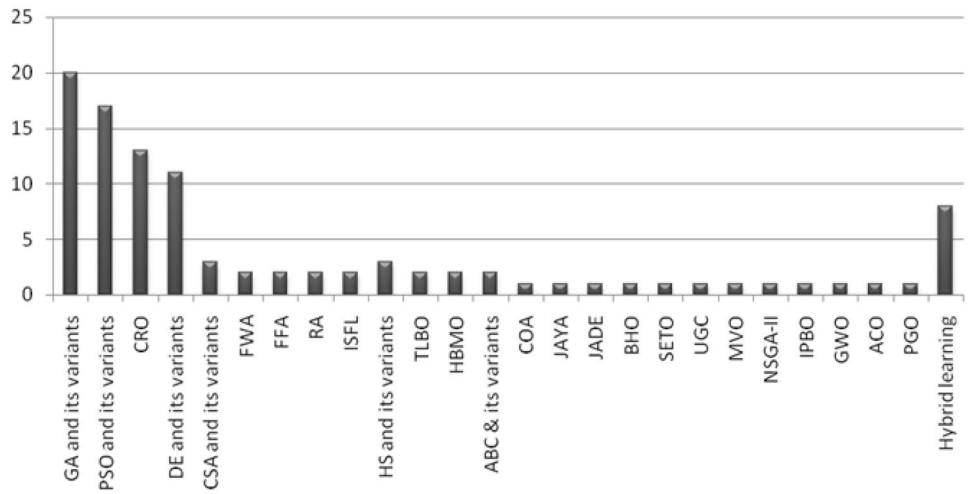


Fig. 22 Contribution of evolutionary algorithms in financial time series prediction

In Table 2, you can see a full evaluation of how well EOAs and ANNs (and, more specifically, HONNs) work together to predict financial time series.

4.1 Technical Discussion

According to the findings of our study (which can be found in Table 2), a variety of evolutionary optimization techniques, in particular those utilized in the process of predicting financial time series, produce reliable results. A pie chart illustrating the relative contribution of various different evolutionary algorithms to the process of financial forecasting is presented for your perusal in Fig. 22. (as far as our survey is concerned).

The review of the relevant literature makes it abundantly clear that the majority of the researchers used genetic algorithms (GA) [36, 66, 94, 100, 115, 117–120, 126, 132, 150–152, 164, 169, 173, 180] for accurate financial time series forecasting. This is the case due to the fact that GA provides optimal global search, subset selection, and near-optimal parameter search. The combination of HONN and GA resulted in improved generalization, a more rapid convergence with global optima, reduced structural complexity in the network, and increased accuracy in predictions. To take the performance of GA one step further, certain upgraded versions of GA, such as multi-objective GA (MOGA) [73] and hybrid GA (HGA) [87], have also been presented and have demonstrated their effectiveness in financial time series forecasting. PSO [78, 107, 131, 138, 149, 157, 162, 165] is a simple algorithm that may be effectively employed to fine-tune the parameters of a wide variety of HONNs due to its lack of complexity, simple parameters, and constraints. Authors have created variations of PSO, such as MPSO [81, 92, 93, 97], HMRPSO [141], RWPSO and WLDPSO [146], and IPSO [76, 108] to address the slow convergence that is sometimes seen.

In order to get better results, a hybrid learning approach called CRPSO was introduced in article [137]. In [134], the network parameters are adjusted using a combination of PSO and a discrete PSO (DPSO) to boost precision. Faster convergence, higher accuracy, fewer control parameters, and efficient optimization of the many HONNs' parameters are all advantages of CRO [38, 72, 77, 90, 113, 114, 124, 125, 127, 171, 177, 181], whereas CRO-LMS [55] outperforms both CRO and LMS learnings due to its hybrid nature. FWA [33, 34] was found to have faster convergence, the ability to find global minima, faster learning, and more accurate predictions than GA, PSO, and GD learning. Improved accuracy in prediction is one of the many benefits of the random search capacity, adaptive stability, and dependable predictability provided by FFA [21, 122, 196]. In contrast, DE [37, 74, 85, 88, 91, 109, 123, 140, 167, 178] is superior to BP, PSO, and HMRPSO when it comes to optimizing the parameters of different HONN because of its ease of use, small parameter set, rapid convergence, and flexibility. Furthermore, a hybrid learning technique known as SDE [68] is effective in locating the best possible solution at a fast convergence rate. With the ABC and DE presented in [178] and a hybrid learning approach ABC-DE presented in [170], there is improved prediction accuracy, greater generalization, a shorter training period, a more balanced search strategy, optimized weights and biases, and shorter training periods.

Additionally, there is a more balanced search strategy. While ACO [178, 198] and a hybrid ACO called genetic ant colony algorithm (GACA) [199] provides a forecast that is both more accurate and ideal, A GWO [155], on the other hand, was able to provide superior ideal outcomes [155]. CSA makes it possible to handle effectively any and all critical parameters [32]. When compared to CSA, hybrid learning strategies such as CMCSA [147] and PGCSA [168] are more capable of determining the parameters of the HONN like the ELM in order to construct hybrid models for financial time series forecasting. It has been discovered that the Rao algorithm, also known as RA, is very effective in searching for the optimal parameters of FLANN [102] and ANN [194] in order to deliver the ideal solution for financial time series forecasting. Better results were achieved by RA in terms of prediction accuracy, computation time, statistical significance testing, and the lowest values for loss functions such as MAPE and ARV values. When compared to BP and DE learning, the error function values obtained using an HS [17] were superior. By doing away with the HS's two constant parameters (bandwidth and pitch adjustment rate), a new variation of the HS known as self-adaptive harmony search (SAHS) [104] has been created to boost the algorithm's performance. SADHS-OELM [130] is a hybrid learning method that has been modified to improve upon the performance of DE, SADHS, and other similar methods by optimizing both the optimum parameters and the network structure (hyper parameters) of a two-hidden-layer feed-forward neural network (SLFN).

On the other hand, ISFL [71, 154] offers advantages over SFL, PSO, and DE in terms of learning speed, optimality of solutions, and global and local exploration. TLBO [14] is parameter less. TLBO improves accuracy. Opposition-based learning with Elitism TLBO (OLE-TLBO) [103] outperformed GA, PSO, TLBO, E-TLBO, and OL-TLBO in parameter optimization. HBMO [75] calculated PSNN parameters better than GA, PSO, and DE. HBMO-GDL version [105] enhanced accuracy and was found superior to GA, PSO, and HBMO. ABC successfully optimizes PSNN parameters [19] for higher accuracy. An improved ABC-DE [175] outperformed DE, PSO, and ABC to train the ELM network. Similarly, other EOAs like COA [69], JAYA [35],

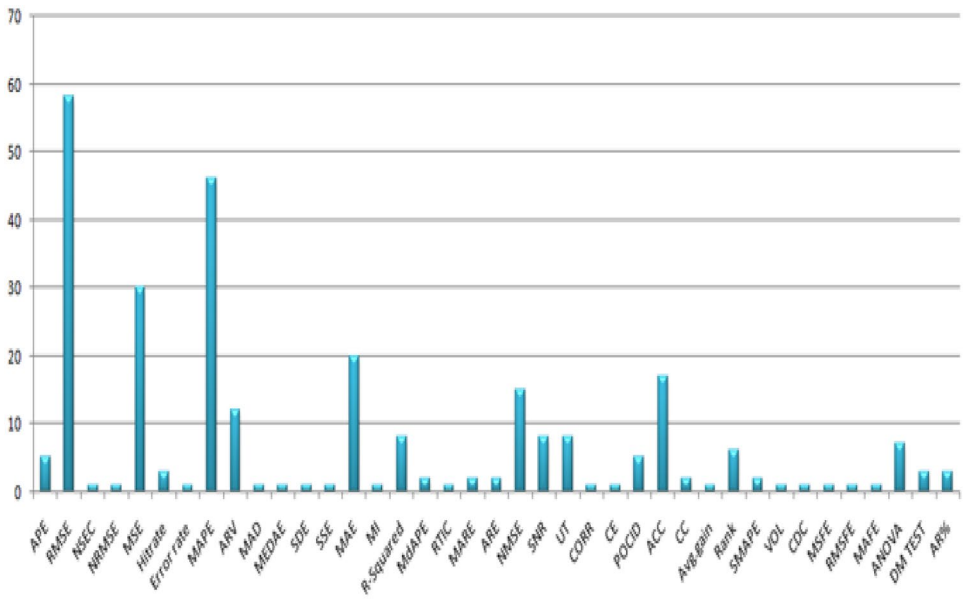


Fig. 23 Performance metrics that are widely used to gauge how effective EOAs-based HONNs

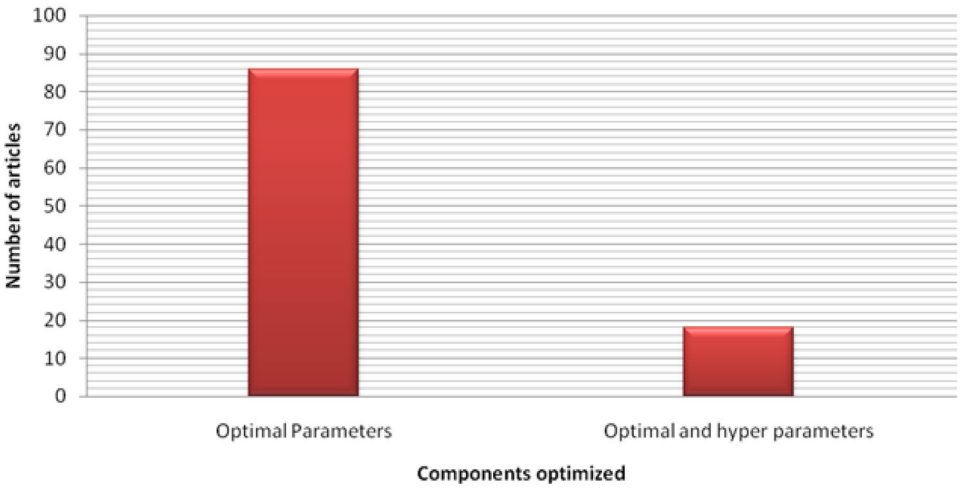
JADE [174], BHO [99], SETO [197], UGC [148], MVO [172], RLS [121], NSGA-II [129], IPBO [179], ACO & PGO [135], PSO-GA [86, 106], ABCDE [170], HMOEA [3], Elitism AEFA [176], and MCS-MCMC [79, 80] also produced effective results when combined with different HONNs for financial prediction.

Financial forecasting involves a wide variety of tasks, such as training the prediction model, optimizing the model's parameters, minimizing the loss function, and increasing the predictor's accuracy. After reading this, you should have a firm grasp on the significance and effectiveness of evolutionary algorithms across these tasks. However, they are well suited for comprehending the data's nonlinearity and require less time to train the neural network or the prediction model to rapidly converge. As a result, we may draw the conclusion that evolutionary algorithms are superior to other methods for training ANNs, in particular HONNs, to the state-of-the-art level, enabling precise financial forecasting **(RBQ2).** That's why it makes sense to combine evolutionary algorithms and HONNs to provide accurate forecasts of financial time series.

4.2 Performance Indicators for EOAs based HONNs

Our review of the research shows that several researchers have used different statistical tools and performance measures to evaluate how well the proposed HONN model and EOAs work. They are as follows: average percentage error (APE) [66, 90, 111, 112, 119], root mean square error (RMSE) [3, 14, 17, 19, 21, 32, 36, 37, 68, 71–79, 81, 84–86, 88, 90, 92, 93, 99, 103, 104, 106, 109, 110, 112, 120, 122, 123, 130, 131, 135, 138, 140, 141, 149, 153, 154, 156, 157, 159, 161–164, 164, 165, 167, 175, 178, 180, 200], nash–sutcliffe efficiency coefficient (NSEC) [178], normalized Root Mean Square Error (NRMSE) [163], mean squared error (MSE) [3, 17, 32, 68, 71, 80, 82, 86–89, 96, 101, 109, 110, 116, 122, 129, 132, 137, 138, 146, 147, 155, 157, 165, 168–170, 174, 179], Hit rate [112, 115, 131], Error rate [131], mean absolute percentage error (MAPE) [34, 36–38, 55, 67–69, 71, 81, 85, 92–94, 102, 109, 110, 118, 121–126, 128–130, 136–141, 146, 147, 150, 154, 162, 165, 167, 168, 170–173, 176, 180, 181, 200], average relative variance (ARV) [38, 67, 94, 121, 124–126, 129, 134, 150, 171, 180], mean absolute deviation (MAD) [110], median absolute error(MedAE) [73], SDE [55], SSE [55], mean absolute error (MAE) [19, 32, 36, 71, 73, 82, 88, 89, 96, 101, 135, 146, 147, 151, 152, 154, 161, 168, 169, 175, 200], MI [135], R-Squared [36, 71, 110, 124, 125, 139, 153, 171], median absolute percentage error (MdAPE) [162, 171], RTIC [146], mean absolute relative error (MARE) [71, 175], average relative error (ARE) [164], normalised mean squared error (NMSE) [3, 82, 84, 89, 96, 101, 126, 135, 142–144, 157–161, 171, 201], Signal to noise ratio (SNR) [82, 89, 95, 96, 101, 107, 160, 161], UT [67, 68, 71, 121, 126, 129, 150, 180], correlations (CORR) [147], CE [146], prediction of change in direction (POCID) [94, 125, 150, 180, 181], accuracy (ACC) [14, 21, 75, 77–79, 86, 98, 99, 104–106, 108, 129, 131, 174, 178], correlation coefficients (CC) [68, 156], avg.gain [35], rank [72, 79, 110, 115, 117, 172], symmetric mean absolute percentage error (SMAPE) [72, 74], ANOVA [77, 78, 86, 100, 105, 106, 134], deibold-mariano test (DM Test) [33, 38, 124], annual return percentage (AR%) [142–144], volume (VOL) [142], correct directional change (CDC) [142], mean squared forecast error (MSFE) [130], root mean square forecast

Fig. 24 Components optimized



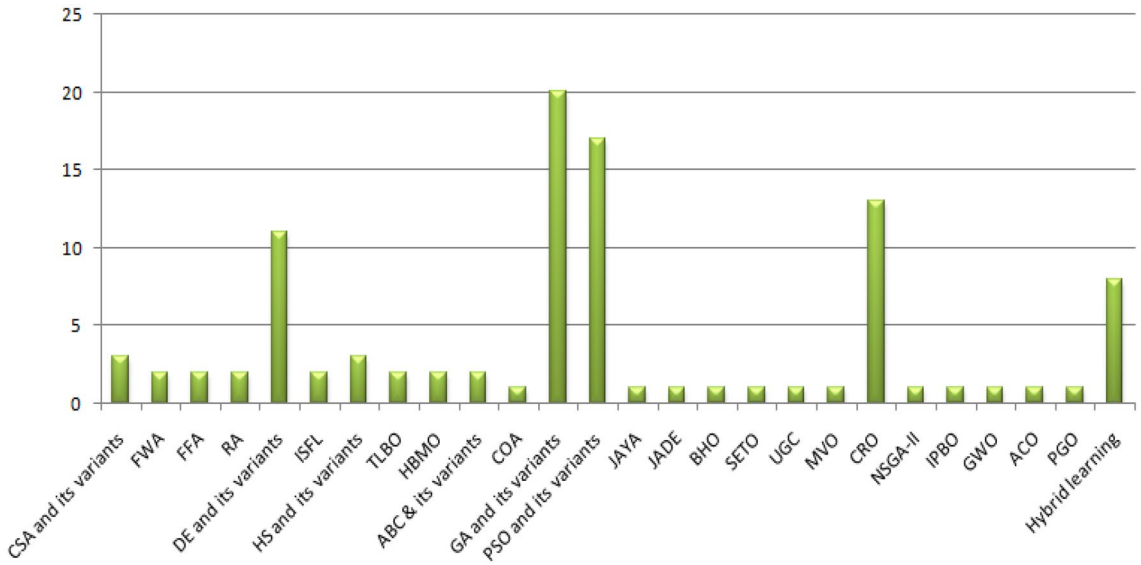


Fig. 25 Contribution of EOAs in Optimization of HONNs components

error (RMSFE) [130], and mean absolute forecast error (MAFE) [130]. The appropriate formulas for some of the well-known performance indicators are given in the Appendix section. Figure 23 shows a bar chart with several performance metrics that are widely used to gauge how effective EOAs-based HONNs are at predicting the financial time series.

4.3 Optimization of HONN Components

Evolutionary techniques optimise HONN component estimation (such as hyper-parameters, weights and biases, hidden layer size, hidden layer neurons, learning rate, activation function, epochs, population size, etc.). HONN designs employ several evolutionary methods to optimise components (according to the literature in Table 2). Figure 24 shows a bar chart view of frequent optimizations of different components of HONNs as far as our review is concerned.

On other hand, Fig. 25 represents the frequent participation of evolutionary algorithms in optimizing the components of HONNs.

4.4 Discussion and Statistical Results

Table 1 shows how many HONNs are used for predicting financial time series, and Table 2 shows how academics have tackled HONN optimization utilizing a variety of EOAs in the past and present to boost prediction accuracy. The optimization of all essential factors, however, requires more

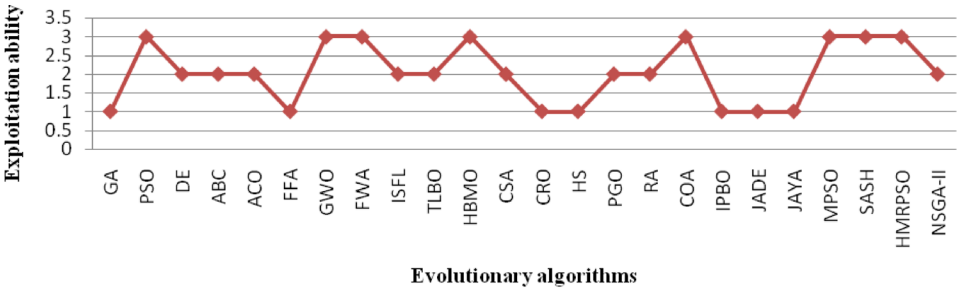
Table 3 Components of EOA in different criteria

EOA	Exploitation ability	Exploration ability	Convergence ability	Global minima ability	Parameter setting	Generalization ability
GA	Medium	High	Slow	Low	Medium	Medium
PSO	Very high	Low	Fast	Low	Medium	Easy
DE	High	Medium	Very fast	High	Easy	Easy
ABC	High	Medium	Moderate	Medium	Easy	Easy
ACO	High	Medium	Fast	High	Hard	Easy
FFA	Medium	Low	Moderate	Low	Easy	Easy
GWO	Very high	High	Very fast	Very high	Easy	Medium
FWA	Very high	High	Very fast	Very high	Easy	Easy
ISFL	High	Medium	Fast	High	Medium	Easy
TLBO	High	High	Moderate	Medium	Medium	Low
HBMO	Very high	High	Fast	Very high	Easy	Low
CSA	High	Very high	Moderate	Very high	Easy	Easy
CRO	Medium	High	Moderate	High	Easy	Easy
HS	Medium	High	Fast	High	Easy	Easy
PGO	High	Medium	Moderate	Medium	Easy	Easy
RA	High	Very high highhigh	Fast	Very high	Easy	Easy
COA	Very high	Very high	Very fast	Very high	Medium	Easy
IPBO	Medium	High	Very fast	Very high	Medium	Medium
JADE	Medium	Medium	Fast	High	Medium	Medium
JAYA	Medium	High	Fast	High	–	Easy
MPSO	Very high	Medium	Fast	High	Medium	Easy
SASH	Very high	High	Fast	Very high	Medium	Easy
HMRPSO	Very high	High	Very fast	Very high	Medium	Medium
NSGA-II	High	High	Fast	High	Hard	Medium

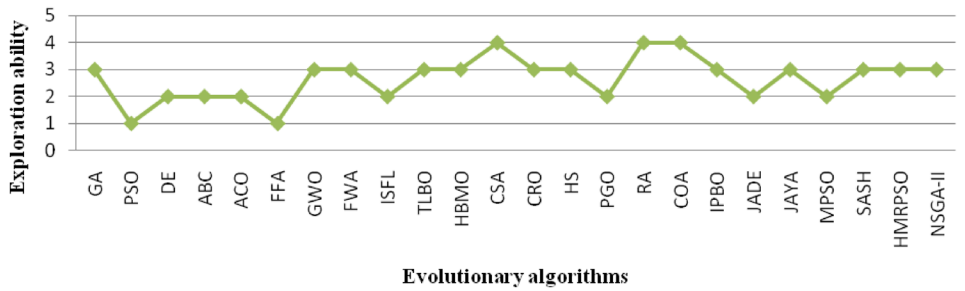
research. Evolutionary approaches are frequently employed to optimize HONN designs for performance. According to the published works, the HONN was taught with both well-known and recently suggested EOAs. Yet, the No Free Lunch (NFL) hypothesis asserts that every situation is different, and hence many methods must be used to effectively anticipate financial time series. According to the NFL theorem, developing a single evolutionary algorithm that can solve many different types of problems is nearly impossible. Hence, it seems that an evolutionary algorithm might not be the best choice for improving the HONN parameters. The fact remains, nevertheless, that it works well for fixing some problems. The only way to tell if the evolutionary algorithm has converged is through experimental assessments. Due to the nature of evolutionary algorithms (which explore all possible solutions to a problem via the operators they utilize), knowing whether an evolutionary algorithm is the best strategy can be challenging. It is for this reason that non-trivial methods must be used in order to get optimal values for the HONN parameters. Improvements to typical implementations of EOAs and their combinations have been identified in a large number of optimization studies to boost their exploitation and exploration power.

New evolutionary algorithms have been developed as a result of recent research, and they have already shown themselves to be superior to their forerunners in a number of optimization scenarios. This presents a fresh challenge for developing both new and existing evolutionary methods to optimize the parameters of HONN for effective time series forecasting, as shown in Table 2 of current works. The vastness of the search space presents a significant obstacle for large-scale optimization efforts. In addition, the algorithm's convergence is rather insensitive to modifications to its parameters. As a result, even if the researchers have carefully computed the algorithm's initial parameters, it will not be able to effectively investigate and make use of a large dataset of great complexity.

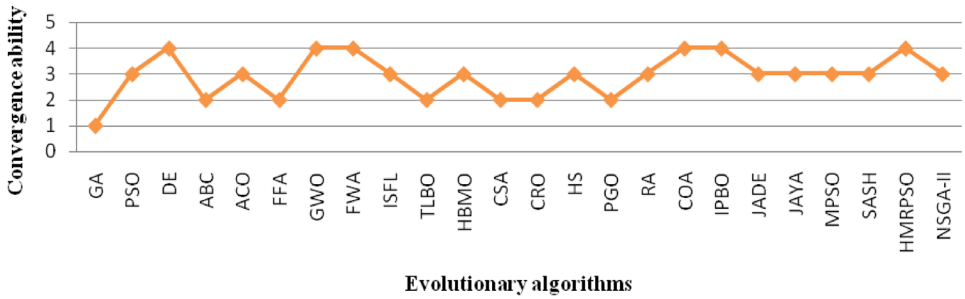
To achieve global and local search coverage, we need to use robust operators that enhance both exploration and exploitation. This problem may be solved with evolutionary algorithms by borrowing their advantages and operators to solve new problems. Over the past decade, the academic community has been using hybrid algorithms to improve



(a): Exploitation ability of evolutionary optimization algorithms



(b): Exploitation ability of evolutionary optimization algorithms



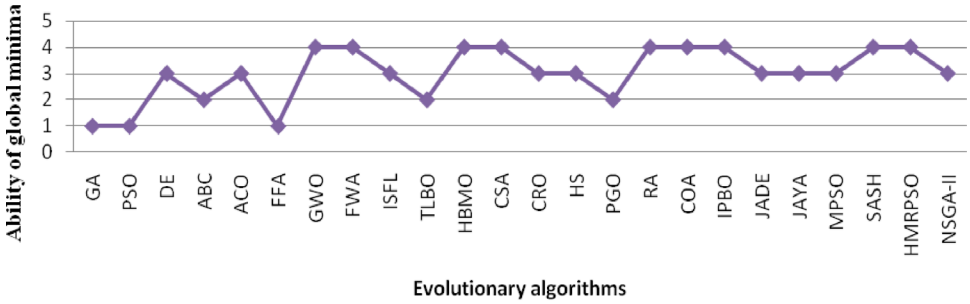
(c): Convergence ability of evolutionary optimization algorithms

Fig. 26 **a** Exploitation ability of evolutionary optimization algorithms **b** Exploitation ability of evolutionary optimization algorithms **c** Convergence ability of evolutionary optimization algorithms

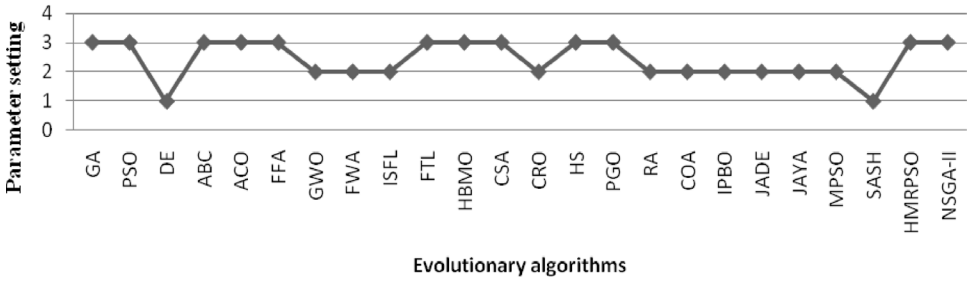
optimization performance. One algorithm's shortcomings may be reduced by employing a different algorithm. The optimum evolutionary performance may be attained by a healthy equilibrium of exploration and exploitation capacities. To avoid becoming stuck in local minimums and to explore the search space more effectively, the exploration operator can do a global search; however, this may result in a slow convergence. It's true that the abuse operator results in very high rates of convergence, but if it attempts to move too quickly, it may get stuck. Existing evolutionary algorithms vary greatly in their ability to either avoid or take advantage of local optimums and/or convergence trends. In Table 3, you may compare how well different EOAs perform in terms of their global optimum ability, convergence trend, exploitation power, exploration power, parameter settings, and generalization ability. For improved outcomes while using the HONN, the operators of BHO, ISFL, DE, FWA, FFA, CRO, CSA, RA, MPSO, HS, PGO, JAYA, JADE, and GWO also performed well. This strategy has the potential to aid researchers in developing superior hybrid algorithms for the forward-looking prediction of financial time series.

Data in Table 2 reveals that little research has focused on optimizing HONN hyperparameters as opposed to the more common practice of identifying optimal HONN parameters. This will help researchers improve the network as a whole in the long run.

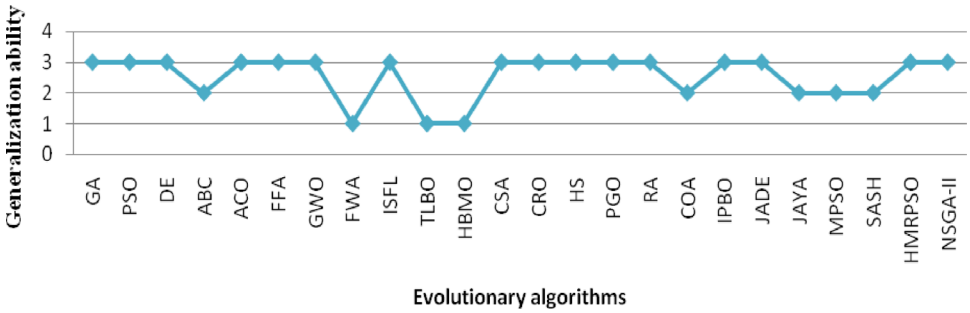
To plot EOA's abilities, we translated categorical ratings into numerical values: extremely high = "3", high = "2" and medium = "1" to plot EOA's exploitation capabilities. Very high signifies "4", high means "3," medium means "2," and low is "1" for EOA's exploration capabilities. To plot EOA's convergence ability, we used "4" for very fast, "3" for fast, "2" for moderate, and "1" for low. Figure 26 a–c shows the



(a): Ability of global minima of evolutionary optimization algorithms



(b): Parameters setting of evolutionary optimization algorithms



(c): Generalization ability of evolutionary optimization algorithms

Fig. 27 **a** Ability of global minima of evolutionary optimization algorithms **b** Parameters setting of evolutionary optimization algorithms **c** Generalization ability of evolutionary optimization algorithms

graphs for exploitation, exploration, and convergence for both well-known EOAs and new ones.

We translated categorical data like "very high" = 4, "high" = 3, "medium" = 2, and "low" = 1 to plot EOA's global minima ability. We mapped category values like "hard" = 3, "medium" = 2, and "easy" = 1 for parameter settings. For EOA's generalization ability, "mid" = 3, "easy" = 2, and "poor" = 1. EOA's global minima, parameter settings, and generalization ability are depicted in Figs. 27 a, b, c.

5 Conclusion and Future Scope

Even though it's difficult to make accurate projections of financial time series, HONN has made strides in this direction. HONN and its variations perform exceptionally well in economic forecasting because of their high-order terms and relatively straightforward structure. HONN has been shown to be a highly effective model for financial classification and prediction since it does not have gradient issues and hence does not converge to a local minimum. Overfitting and under-fitting are not likely to occur because there is just one layer hiding beneath the surface. HONNs are well suited to revealing the non-linearity and concealed elements of financial data due to their exceptional mapping power. With the addition of higher-order expressions, HONNs have proven to be effective in tackling complex real-world problems, especially in financial time series forecasting. As we've shown in our studies, HONN and its variants produce excellent results when used to forecast financial time series. It is demonstrated that HONN excels in several ANN-related metrics, including accuracy, loss function, convergence, learning rate, and adaptability.

Some of the best reasons to use HONNs to predict future values in financial time series are listed below:

- Since they have fewer nodes, lower-order networks are simpler.
- They are highly adaptable and have good generalization ability.
- They're lightweight since there's only one hidden layer.
- A tendency that can tolerate error.
- HONNs are convenient to train because of their rapid learning speed.
- They have high capabilities for approximation and very effective non-linear mapping capabilities.

Nevertheless, hybrid models that mix EOAs and HONNs have been shown to be the most successful for financial time series forecasting. Many comparisons have been made between EOAs-based and gradient-based HONNs, and we found that researchers have employed a broad variety of EOAs to establish values for the HONNs' optimum and hyperparameters. To cap it all off, HONNs trained with EOAs show superior performance. Similarly, some researchers have discovered that, when training HONNs for financial time series forecasting, certain hybrid learning approaches perform very well in comparison to conventional EOAs. Also, it was discovered that some recently proposed EOAs are superior to both GD learning and some classic EOAs. Some of the best reasons to utilize EOAs to anticipate future values in financial time series are stated below:

- Because of their greater versatility in determining both the optimum and hyperparameters of the network model,
- EOAs are more suited to train ANNs since they have greater exploration and exploitation capabilities.
- They are capable of producing better convergence.
- They are capable enough to perform global search.
- They are not affected by the local minima problem.
- They are powerful enough to detect nonlinearity in time series datasets.

In this thorough review, we found that HONNs are good enough for accurately modeling the hidden dynamics of financial time series, and EOAs are better than traditional back propagation methods for adjusting the HONNs' parameters. We discovered the efficacy of EOA-based HONNs in creating hybrid models with improved prediction performance in this study. So, we have been greatly encouraged by the performance of HONNs in financial time series forecasting as well as by the helpful contribution of EOAs when hybridized with HONNs in the same forecasting domain. In order to better anticipate financial time series, we want to use a future hybrid model that combines the most effective HONNs with a recently suggested EOA.

In addition to finding a new EOA to train the optimum parameters and hyperparameters of the HONNs to predict the financial time series one step ahead of time, interested researchers may also find ways to enhance the loss function and deliver maximum accuracy in financial time series forecasting. To optimize the HONNs for predicting financial time series, a number of hybrid learning methods have been presented and proven to be more effective than either EOA or GD-based learning alone. However, the most effective hybrid method of education has yet to be discovered. Interested researchers could create novel hybrid learning by developing and deploying some other EOA-based hybrid learning to train HONNs for financial time series prediction. Readers and researchers interested in expanding this work's future potential may find it useful to combine deep reinforcement learning [50] with clustering [51] and EOAs to develop a novel hybrid model for predicting the future value of financial time series.

Appendix

$$\text{RMSE} = \sqrt{\sum_{i=1}^{N} \frac{(\hat{Y}_{\text{Pr}edicted} - Y_{Observed})^2}{N}}$$

$$\text{MAE} = \frac{\sum_{\text{I}=1}^{\text{N}} \left|\hat{Y}_{\text{Pr}edicted} - Y_{Observed}\right|}{N}$$

$$\text{M}SE = \frac{1}{N}\sum_{I=1}^{N} (Y_{Observed} - \hat{Y}_{\text{Pr}edicted})^2$$

$$\text{RME} = \frac{1}{N}\sum_{I=1}^{N} \frac{\left|Y_{Observed} - \hat{Y}_{\text{Pr}edicted}\right|}{\left|\hat{Y}_{\text{Pr}edicted}\right|} \times 100\%$$

$$R^2 = 1 - \frac{\sum_i (Y_{Observed} - \hat{Y}_{\text{Pr}edicted})^2}{\sum_i (Y_{Observed} - \overline{Y}_{\text{Pr}edicted})^2}$$

$$\text{M}APE = 100\% \times \frac{1}{N}\sum_{i=1}^{N} \frac{\left|Y_{Observed} - \hat{Y}_{\text{Pr}edicted}\right|}{Y_{Observed}}$$

$$\text{U of Theil} = \frac{\sum_{i=1}^{N} (Y_{Observed} - \hat{Y}_{\text{Pr}edicted})^2}{\sum_{\text{i}=1}^{\text{N}} (Y_{Observed} - Y_{Observed+1})^2}$$

$$MAD = \frac{1}{N}\sum_{i=1}^{N}\left|d_i - mean(D)\right|$$

$$\text{Coefficient of Variation (CoV)} = \left(\frac{S}{M}\right) \times 100,$$

S represents standard deviation and M stands for Mean

$$\text{Corelation (CORR)} = \frac{\sum_{i=1}^{N_{test}} (A_i - mean(A_i))(P_i - mean(P_i))}{\sqrt{\sum_{i=1}^{N_{test}} (A_i - mean(A_i))^2 \sum_{i=1}^{N_{test}} (P_i - mean(P_i))^2}}$$

$$MAAPE = \frac{1}{N_{test}}\sum_{i=1}^{N_{test}}\left(AAPE_i\right) \text{ Where, AAPE}_i = \arctan\left(\left|\frac{V_{actual} - V_{predicted}}{V_{actual}}\right|\right)$$

$$\text{POCID} = 100 \times \left(\frac{\sum_{i=1}^{N} D_i}{N}\right) \text{ where } D_i = \left\{\begin{array}{l} 1, \text{ if } (Y_{Observed} - Y_{Observed-1})(\hat{Y}_{Predicted} - \hat{Y}_{predicted-1}) > 0 \\ 0, \text{ otherwise} \end{array}\right\}$$

In the above formulas, N represents total number of observations, $Y_{Observed}$ represents observed output, $\hat{Y}_{predicted}$ represents predicted output, and $\overline{Y}_{predicted}$ is the mean of the predicted output.

$$\text{SMAPE} = \frac{100\%}{N}\sum_{t=1}^{N}\frac{\left|Forcastvalue_t - Actualvalue_t\right|}{\left(\left|Actualvalue_t + Forecastvalue_t\right|\right)/2}$$

$$NRMSE = \frac{\sum (Simulatedvalues_i - Observedvalues_i)^2}{\sum (Observedvalues)^2}$$

$$ARV = \frac{\sum_{i=1}^{N}(E_i - A_i)^2}{\sum_{i=1}^{N}(E_i - M)^2}$$ Here E_i is the estimated value, A_i is the Actual value, M is the mean of datasets and N is the total number of datasets.

Funding This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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